

From Descriptive Atlas to Tradeable Signal:

An Out-of-Sample Test of the Multi-Asset Conditional Exceedance Framework as a Portfolio Tilt Strategy

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Abstract

This paper is the third in a series built around the Conditional Probability of Exceedance (CPE) framework — a live, production-deployed signal system covering 161 instruments across equities, FX, crypto, volatility, commodities and rates. The first paper (Ramanathan, 2026b) characterised conditional tail co-movement descriptively across that universe. The second (Ramanathan, 2026a) tested a binary-gate trading rule on SPY alone; it found no edge, tracing the failure to the gate discarding the shape of the underlying probability surface. This paper closes that gap: it builds a multi-predictor, continuously-weighted Portfolio Tilt strategy across five tradeable asset-class sleeves (Equities, Gold, Bonds, Crypto, FX), with all thresholds and joint configurations frozen on data through 2024-12-31 and evaluated walking forward through all 250 trading days of 2025. The five-sleeve strategy (Sharpe 0.560) did not outperform its own neutral-weight benchmark (Sharpe 0.563), with the sole signal episode traced to structurally decaying volatility-complex ETPs whose long-horizon quantiles are non-stationary — a confound diagnosed and disclosed; the positive results in Sections 14–18 follow a different methodological path (economic prior and episode-conviction sizing) rather than a direct fix of the decay mechanism. The key positive finding emerges from the corrected specification: an episode-independence-aware, conviction-sized cross-sectional strategy across 61 economically validated targets achieves hold-to-horizon Sharpe 1.224 (pct_exceeding 1.8%, statistically significant) in 2025, with the best cross-sectional structure reaching Sharpe 1.613 (pct_exceeding 2.2%) — outperforming TSMOM (0.822), XSMOM (0.757), trend-following EWMA (0.956) and risk parity (1.555). These are the first statistically significant results in the three-paper series. The central data-sufficiency caveat, established in Section 20, is that the validated channel's predictors (VIXM, VIXY) have only four stored independent training-period episodes across fourteen years of history; whether the 2025 result reflects genuine predictive content or the minimum episode history required to produce a selective screen by chance cannot be determined from the current record. A further three to five independent conditioning episodes — likely requiring a decade or more of additional data — would be needed to distinguish these hypotheses. The central methodological finding is that no purely statistical filter — shrinkage, sample-size weighting, or episode-independence counting — can substitute for an economic prior over admissible predictor-target relationships; the paper provides both the diagnostic evidence for this claim and a concrete implementation of such a prior (~91 admissible channels, four-tier confidence system). A partial 2026 window (~5 months, 104 trading days) is reported for transparency only; it is not a second evaluation. Section 20 reports a proper out-of-sample walk-forward test in which the full CPE pipeline is re-derived from pre-evaluation data across 2010–2024 (15 evaluation years), with a single-screen continuous evaluation over 2015–2025; these tests establish that the framework's 2025 result is training-window dependent, that the validated vol→equity channel requires a minimum of four stored independent training-period episodes to dominate a selective screen (Section 20.6 establishes that the stored n_{episodes} for the validated configurations is 4, correcting an earlier runtime diagnostic), and that the CPE framework functions as a rare-event detector whose signal quality is contingent on sufficient episode history in the channel being tested.

1. Introduction and Relation to Prior Work

Two earlier studies established the building blocks this paper combines. The Conditional Exceedance Atlas (Ramanathan, 2026b) computed the Conditional Probability of Exceedance (CPE) descriptively across a 161-instrument, six-asset-class universe, including a greedy joint-conditioning procedure that aggregates up to ten simultaneously-firing predictors into a single joint CPE. No threshold in that study was held out of sample, and no trading claim was made. A separate single-asset study (Ramanathan, 2026a) took the narrower, single-predictor version of the same construct and subjected it to a strict pre-2025/2025 train-test split on SPY alone; the descriptive lift was robust (mean 2.56× across 54 configurations) but a mechanical binary entry rule built on top of it did not produce a demonstrated edge, and the gap was traced specifically to the rule discarding the shape of the underlying conditional probability curve in favour of a single eligibility gate.

That single-asset paper's own Section 8.15 flagged the natural next test directly: whether a structurally richer construction — multi-predictor, continuously-weighted, aggregated across an asset-class book rather than collapsed to one binary fire/no-fire decision on one instrument — performs differently. This paper is that test. It is deliberately not a re-run of the single-asset paper's binary entry-rule logic at greater scale; it instead adapts the Portfolio Tilt construction already used in the author's live (in-sample) dashboard, but rebuilds it under the same train/test discipline established in the single-asset paper: every threshold, every joint-configuration selection, and every neutral portfolio weight is computed exclusively from data through 2024-12-31, and the resulting frozen rule is then evaluated by walking forward, day by day, through the actual 2025 price path — a true out-of-sample test, not an in-sample snapshot of “what the dashboard would say today.”

2. Methodology

2.1 Train/Test Split and Leakage Control

All thresholds, all pairwise and joint CPE configurations, and all neutral-weight Sharpe ratios are computed using price data dated 2024-12-31 or earlier. This is enforced programmatically: every script that touches the training pipeline truncates the price panel before any computation begins and asserts that the resulting maximum date does not exceed the training cutoff. The 2025 evaluation walk-forward script loads the full price history (through 2025-12-31 only — no 2026 data enters this paper at any point) purely to compute each day's live trailing increments, which are then compared against the frozen training-period thresholds; the thresholds themselves are never re-estimated using 2025 data.

This discipline matters because the single-asset paper's own Section 6.1 documents a real instance of this exact leak (an entry-signal threshold computed on the full price series rather than the training period alone) being caught and corrected after the fact. During development of this paper's pairwise CPE engine, an analogous but distinct bug was caught by independent verification: a vectorised implementation initially counted conditioning observations at dates within the training window where a predictor fired but the target's forward outcome fell after the training cutoff and was therefore unobservable. Restricting every conditioning count to dates where the forward outcome is itself inside the training window (not just where the predictor's own trailing window is) corrected this; the fix was verified by reproducing sixteen independently-selected (Y, X, horizon, threshold, direction) cells against a separate loop-based reference implementation, with sixteen exact matches.

2.2 Pairwise and Joint CPE Screen (Train-Only)

The pairwise screen follows the Atlas's own definition (Equations 1–5 of that paper) exactly, restricted to the training window: for each ordered pair of instruments (Y, X), trailing horizon τ_p , forward horizon τ_f , and quantile pair (q_x, q_y), the conditional probability of a forward exceedance of Y given a trailing exceedance of X is estimated as a relative frequency over the training sample, with both bullish (upper-tail) and bearish (lower-tail) directions evaluated. A configuration survives only if $CPE \geq 0.80$, $lift \geq 1.50$, and the conditioning sample contains at least 100 training-period observations — the same filter bounds used throughout both prior papers.

To keep the full 161-instrument, six-asset-class sweep computationally tractable in this environment, the parameter grid used here is coarser than the Atlas's full grid: trailing horizons $\tau_p \in \{5, 21, 63, 126, 252\}$, forward horizons $\tau_f \in \{21, 63, 126, 252\}$, and quantile levels $q \in \{0.70, 0.75, 0.80, 0.90, 0.95\}$ (both tails). This is a coarser grid than the Atlas's full sweep but matches the horizons already used in the author's live Portfolio Tilt dashboard, and the resulting pairwise screen still surveys $161 \times 158 \times 5 \times 4 \times 5^2 \times 2 \approx 51$ million candidate configurations, exclusively on training data. 14,707 configurations survived the filters.

The joint screen applies the Atlas's own greedy intersection procedure (its Section 3.2) unchanged: for each (Y, τ_f , q_y , direction) group, start from the single best pairwise predictor, then repeatedly add whichever remaining (unique-ticker) candidate most increases the joint CPE subject to the same survival filters, stopping at six predictors or when no admissible addition exists. 1,655 joint configurations of size 2–6 survived; their distribution across set sizes (662 / 446 / 347 / 112 / 88 for sizes 2 through 6) is consistent with the same data-thinning pattern documented in the Atlas's Section 5.1, where larger predictor sets necessarily draw on a shrinking pool of overlapping training-period episodes.

2.3 From Joint Configurations to a Continuous Tilt Score

This is the structural departure from the single-asset paper's binary entry rule, and the part of the design this paper exists to test. Rather than thresholding a single CPE value into a fire/no-fire decision, every surviving joint configuration Π is assigned a quality weight

$$w(\Pi) = CPE(\Pi) \times lift(\Pi) \times \ln(n_{joint}(\Pi)) \times h(\Pi)$$

identical to the Atlas's own signal-score weighting (Section 6.1), with one addition: $h(\Pi)$, a mild history-length down-weight applied per joint configuration. $h(\Pi)$ is the minimum, across every predictor in the set, of a per-predictor weight that ramps linearly from 0.35 at 100 training-period observations to 1.0 at 756 observations (≈ 3 trading years), and is held at 1.0 above that. This directly addresses the short-history confound the Atlas's own Section 6.1 flags explicitly for the spot Bitcoin ETFs (IBIT, FBTC, BITB launched January 2024, leaving only ≈ 245 training-period observations before the 2025 cutoff): rather than excluding these predictors outright — which would discard real information — their contribution to the tilt score is informationally discounted, receiving roughly half weight (0.494) relative to a fully-seasoned predictor.

On each trading day of 2025, every joint configuration's predictor set is checked against the frozen training-period thresholds using that day's live trailing increments. A configuration “fires” on day t if every predictor in its set simultaneously clears its own threshold on day t . Per asset class and per horizon, the day's score is

$$S(class, t) = [\sum w \cdot fire(bullish) - \sum w \cdot fire(bearish)] / [\sum w(bullish) + \sum w(bearish)]$$

normalised to $[-1, +1]$, following the Atlas's own convention exactly. Horizon scores are combined into an overall daily class score using the same horizon weights as the live dashboard (21d: 20%, 63d: 30%, 126d: 30%, 252d: 20%), and mapped to a tilt delta via five symmetric tiers around zero (± 15 percentage points at $|\text{score}| \geq 0.30$, ± 8 at $|\text{score}| \geq 0.05$, 0 otherwise).

2.4 Position Sizing and Tradeable Universe

Each of five asset classes — Equities, Gold, Bonds, Crypto, and FX — trades through a single liquid proxy (SPY, GC=F, TLT, BTC-USD, UUP respectively), matching the convention already established in the author's live Portfolio Tilt dashboard. The volatility complex, while extensively used as a predictor throughout (it is, as shown below, the dominant predictor class for the one signal that did fire), is not itself a tradeable sleeve: VIX-complex products carry severe structural roll-decay that makes buy-and-hold-style capital accounting unsuited to them, and this exclusion was a deliberate scoping decision made before any results were observed.

Neutral (no-tilt) weights are derived from training-period Sharpe ratios exactly as in the live dashboard, but computed using only data through 2024-12-31 rather than a rolling window that would include 2025: Equities 30.87%, Gold 24.29%, Crypto 30.10%, FX 10.00% (fixed), and Bonds 4.74% (floored at the 5% minimum, reflecting a negative training-period Sharpe of -0.075 driven by the 2022–2024 bond bear market within the ten-year lookback). Each day's raw weight is neutral weight plus tilt delta, clipped to $[0, 50]\%$, and renormalised to sum to 100% across the five sleeves. To avoid look-ahead, the weight applied to day t 's realised return is the weight computed using information available through day $t-1$ (i.e. the position is lagged one trading day relative to the signal).

This strategy rebalances daily rather than holding fixed-horizon positions, which sidesteps the entry-window/realisation-window distinction that complicated the single-asset paper's Section 8.1 (where 57–63% of trades entered in 2025 only completed their holding period in 2026): every day's portfolio return here is fully realised within calendar 2025 by construction.

3. Out-of-Sample Results

3.1 Headline Performance

Table 1 compares the strategy's capital-weighted 2025 performance against two benchmarks: a neutral-weight buy-and-hold book using the identical five sleeves with no tilt applied at any point, and outright SPY buy-and-hold (the single-asset paper's own comparator).

Portfolio	Total Return	Ann. Return	Ann. Vol	Sharpe
CPE Portfolio Tilt strategy	7.65%	8.60%	15.34%	0.560
Neutral-weight buy-and-hold (no tilt)	7.60%	—	—	0.563
SPY buy-and-hold	18.01%	—	—	0.955

Table 1. 2025 out-of-sample performance, capital-weighted on a \$100,000 notional book, daily rebalanced. All thresholds and joint configurations frozen on data through 2024-12-31.

The strategy and its own no-tilt benchmark are separated by nine basis points of Sharpe and five basis points of return over the full year — a gap small enough that it requires the randomisation test below to even ask whether it is distinguishable from noise. Both diversified portfolios substantially underperformed SPY buy-and-hold on raw return, for a structurally simple reason visible in Table 2: 2025 was an exceptional year for gold specifically, and a five-sleeve book sized

close to its training-period neutral weights necessarily held less in the year's best-performing single asset, and more in its worst, than a concentrated SPY position would have.

Asset Class	Proxy	2025 Return	2025 Sharpe
Gold	GC=F	62.68%	2.489
Equities	SPY	18.01%	0.955
Bonds	TLT	3.96%	0.388
Crypto	BTC-USD	-7.32%	0.023
FX	UUP	-5.79%	-0.770

Table 2. Individual sleeve buy-and-hold performance, 2025 (for context; not a portfolio-level comparison).

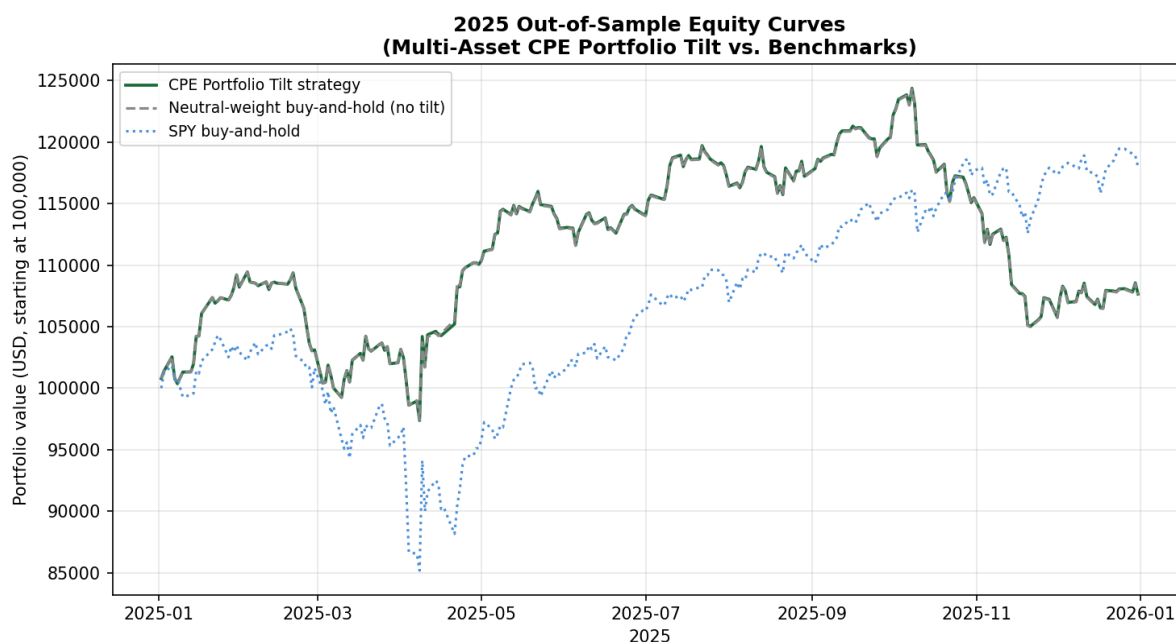


Figure 1. Daily mark-to-market equity curves, 2025. The strategy (green) and its no-tilt benchmark (grey dashed) are visually indistinguishable for almost the entire year, diverging only briefly around the one episode (April 2025) where any sleeve's tilt score moved off neutral.

3.2 Why So Little Fired: The Daily Tilt Record

Figure 2 shows the full daily tilt-score record for all five sleeves across all 250 trading days of 2025. Four of the five sleeves — Gold, Bonds, Crypto, and FX — sat at an exact zero tilt score on every single day of the year. The fifth, Equities, moved only during a roughly six-week window from early April to mid-May 2025, coinciding with the broad volatility spike and recovery around that period's tariff-related equity drawdown, reaching a peak score of 0.39 (comfortably inside the OVERWEIGHT tier) before returning to zero and remaining there for the rest of the year. Across the full 250-day window, only 17 days carried any non-neutral tilt at all, all of them in the Equities sleeve (11 OVERWEIGHT, 6 TILT UP).

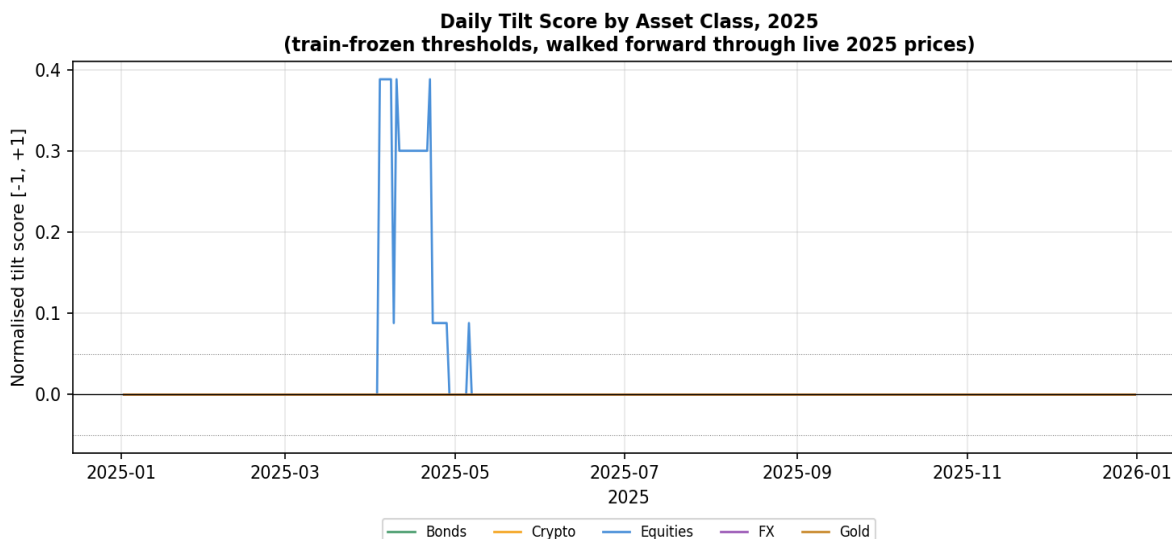


Figure 2. Daily normalised tilt score by asset class, 2025. Four of five sleeves never moved off zero for the entire evaluation year.

This is not a uniform absence of training-period structure — Section 2.2 above and Table 3 below confirm that Gold, Bonds, and FX all have surviving joint configurations on the books, in some cases dozens of them. The absence of firing is a separate, specific finding about why those particular configurations did not re-fire in 2025, examined next.

3.3 The Structural Reason Most Sleeves Never Fired

The most heavily represented joint configurations for the Gold sleeve's bearish signal are built on long-horizon ($\tau_p = 252$ trading days) thresholds of UVXY and VIXY — leveraged and unleveraged short-dated VIX-complex exchange-traded products that the Atlas itself documents (Figure 1b of that paper) as subject to extreme roll decay, spanning “many orders of magnitude over [their] life.” A representative surviving configuration — GC=F bearish, $\tau_f = 252$, conditioned jointly on UVXY and VIXY each clearing their training-period 95th percentile of trailing 252-day log-return — achieved a training-period joint CPE of 1.00 on 129 conditioning episodes (lift 4.0×). The mechanism is structural rather than a defect in the screen: a product that decays persistently over multi-year windows will have a 252-day trailing-return distribution dominated by the secular decay trend itself, so its 95th-percentile threshold sits at an extreme level that is overwhelmingly likely to have been set by the product's own historical decay episodes rather than by a recurring, generalisable regime. Checked against live 2025 prices, neither UVXY's nor VIXY's frozen training-period threshold (log-returns of approximately -307% and -130% respectively, reflecting the decay-dominated 252-day window) was cleared on a single trading day of 2025, so every joint configuration built on them — which constitutes the bulk of the Gold sleeve's bearish signal set — contributed zero firing weight for the entire year.

The one sleeve that did fire — Equities, during the April episode — is itself driven predominantly by VIX-complex predictors, but in the opposite, bullish-on-the-predictor sense: configurations such as SPY bullish at $\tau_f = 252$ conditioned on VXX and VIXM jointly clearing their 90th/95th-percentile trailing 252-day upper-tail thresholds (training joint CPE 1.00, $n = 144$) describe a sustained prior rise in volatility-complex products followed by SPY's subsequent recovery — the same volatility-to-equity channel the Atlas's Section 4.1 identifies as one of its two structurally

dominant cross-class flows. This channel did fire, briefly and correctly, around the one genuine volatility episode the 2025 evaluation window contained.

Asset Class	Bullish configs (any horizon)	Bearish configs (any horizon)	Days non-neutral, 2025
Equities	17	8	17
Gold	3	51	0
Bonds	2	4	0
Crypto	9	0	0
FX	3	24	0

Table 3. Surviving train-period joint configurations per tradeable sleeve, against days the sleeve actually carried a non-neutral tilt in the 2025 walk-forward. Configurations existing on the training-period books did not guarantee any 2025 firing.

3.4 Randomisation Test: Is the Tilt Pattern Distinguishable From Chance?

Following the same logic as the single-asset paper's Section 8.8, the realised daily tilt-delta sequence for each sleeve was independently shuffled across the 250 evaluation days (preserving each sleeve's marginal distribution of tilt values — how many OVERWEIGHT, TILT UP, and NEUTRAL days it had — while randomising which specific days they fell on), the portfolio Sharpe was recomputed for each reassignment, and the resulting null distribution was compared against the actual realised Sharpe. Across 3,000 reassignments, the null distribution had mean 0.564 and standard deviation 0.026; the actual strategy Sharpe of 0.560 was matched or exceeded by 56.3% of random reassignments. By the same convention used in the single-asset paper (a result exceeded by more than roughly 5–10% of random permutations is not distinguishable from chance at any standard significance threshold), the tilt's particular timing pattern in 2025 carries no distinguishable skill over a randomly-timed allocation of the same magnitude and frequency.

3.5 Comparison Against Realistic Peer Strategies, Not Only Buy-and-Hold

Section 3.1's SPY comparison is the same headline benchmark the single-asset paper used, but on its own it understates how the strategy actually performed relative to its true peer group. A 100%-equity buy-and-hold position is not a like-for-like comparator for a five-sleeve diversified tilt strategy in a year when U.S. equities ran hard; the single-asset paper's own Section 8.7 made exactly this point about its binary entry rule, citing SPIVA evidence that the large majority of active strategies also underperformed SPY in 2025. This paper extends that same discipline with comparators closer in construction to a multi-asset tilt strategy specifically, using publicly reported 2025 full-year figures.

Comparator	Construction	2025 Return
This strategy (CPE Portfolio Tilt)	5-sleeve, train-frozen tilt	7.65%
No-tilt neutral-weight benchmark	Same 5 sleeves, fixed weight	7.60%
CTA / managed futures (Simplify CTA ETF)	Systematic trend-following, 50+ futures markets	0.89%
Classic 60/40 (Morningstar US Moderate Target Allocation Index)	60% equities / 40% bonds	≈15%
AQR Risk Parity Fund	Multi-asset risk-balanced allocation	18.86%
SPY buy-and-hold	100% US large-cap equities	18.01%

Gold buy-and-hold	100% gold	62.68%
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Table 3a. 2025 full-year returns for the strategy against a range of publicly reported comparators of varying construction. CTA figure from Simplify Managed Futures Strategy ETF (CTA) 2025 calendar-year return; 60/40 figure as reported by Morningstar ("Markets Brief: A Solid Year for Bonds and 60/40 Portfolios," December 2025); AQR Risk Parity Fund 1-year return as of 12/31/2025 per AQR Funds' published fact sheet. Sharpe ratios are not uniformly available across all comparators from public sources and are therefore omitted from this table; Table 1 above reports Sharpe for the subset where it could be computed directly from price data.

Read this way, the result is more nuanced than Section 3.1 alone suggests, in both directions. Against a genuine systematic-strategy peer — a real, investable trend-following CTA product — this paper's strategy did comfortably better (7.65% versus 0.89%), consistent with the single-asset paper's broader point that 2025 was a difficult year for many systematic approaches, not only this one. But against comparators that are actually closer in spirit to what a five-sleeve diversified tilt strategy is trying to be — a classic 60/40 allocation or a professionally-run risk parity fund — the gap is, if anything, wider than the SPY comparison alone implies: both the 60/40 benchmark ($\approx 15\%$) and AQR's risk parity fund (18.86%) roughly doubled this strategy's return in the same year, for the same broad reason SPY did — 2025 was a strong year for a conventional diversified mix of stocks and bonds specifically (Morningstar reports it was "the best overall year for bonds since 2020"), and a strategy that spent virtually the entire year sitting at fixed, Sharpe-derived neutral weights captured that backdrop no better and no worse than simply holding the same five sleeves with no signal at all. The comparison that matters most for this paper's own research question — does the tilt help, relative to the identical no-tilt portfolio — remains the one already reported in Section 3.1 and 3.4: it does not, distinguishably. The peer comparisons in this section are offered to calibrate that result against the broader investable landscape, not to substitute for it.

4. Interpretation

This paper's own internal logic mirrors the single-asset paper's distinction between a robust descriptive claim and a tested, negative predictive claim, and the two should not be conflated.

The descriptive claim is unaffected by anything in Section 3: 14,707 pairwise and 1,655 joint conditional-exceedance configurations survive permissive but pre-specified filters when estimated on training data alone, and their asset-class distribution (commodities-conditioned-on-crypto as the dense bullish cell, volatility-to-equities as a structurally important bearish channel) is qualitatively consistent with the much larger full-history Atlas screen, despite this paper's coarser parameter grid and strict pre-2025 truncation. The underlying conditional-probability structure the Atlas documents is not an artefact of including 2025–2026 data in its thresholds.

The predictive claim tested here — that aggregating that structure into a continuously-weighted, multi-predictor Portfolio Tilt construction, rather than a single-predictor binary gate, would produce a demonstrated trading edge — does not have a positive answer in this one evaluation year, for a specific and identifiable reason: most of the training-period joint configurations selected by the greedy procedure for the non-Equities sleeves happen to be anchored on structurally decaying instruments whose extreme-quantile thresholds are unlikely to be re-cleared by ordinary subsequent price action regardless of the true underlying market regime. This is a different failure mode from the one the single-asset paper identified (a binary gate discarding the shape of an otherwise-informative probability curve); here, the richer multi-predictor aggregation machinery is in place and does discriminate by magnitude and confidence exactly as designed, but the predictor universe it draws its strongest configurations from is, for several sleeves, dominated by instruments whose long-horizon quantiles are not stationary in the sense the framework's

threshold methodology assumes — precisely the caveat the Atlas's own Section 2.1 and Section 6.1 flag in the abstract, now observed concretely in a live one-year forward test rather than as a theoretical concern.

Section 5 below reports a direct test of the most obvious such fix — restricting the joint-configuration screen to exclude structurally decaying, short-dated leveraged volatility products as predictor candidates, while retaining the VIX-family indices themselves ($\hat{VIX}$, $\hat{VXN}$, $\hat{OVX}$, etc.), which do not carry the same decay mechanism. That test is reported as a single, disclosed, post-hoc specification change, exactly the discipline both prior papers in this series have insisted on throughout, rather than as a parameter sweep performed until a more favourable result emerged.

5. Follow-Up Test: Excluding Structurally-Decaying Volatility Products From the Predictor Pool

Section 4 raised, but deliberately did not test, a direct hypothesis arising from the Section 3.3 diagnosis: would excluding short-dated, roll-decaying VIX-complex exchange-traded products (UVXY, SVXY, VXX, VIXY, VIXM) from the predictor pool — while retaining the VIX-family index levels themselves ($\hat{VIX}$, $\hat{VXN}$, $\hat{OVX}$, $\hat{GVZ}$, $\hat{EVZ}$, $\hat{VVIX}$, $\hat{SKEW}$), which carry no roll-decay mechanism — change the 2025 result? This section reports that test, run as a single, disclosed, post-hoc specification change with every other element of the pipeline (parameter grid, filter thresholds, history-weight ramp, neutral weights, horizon weights, tilt tiers) held identical to Sections 2–3. The exclusion is stated as a hypothesis and then tested fresh against the same 2025 evaluation window, not tuned against it.

5.1 A Ruled-Out Alternative: Short Predictor History

Before reporting the decaying-ETP test, one further natural hypothesis is worth checking and ruling out directly: could the weak 2025 result instead be attributable to predictors with too little training-period history, particularly the spot Bitcoin ETFs (IBIT, FBTC, BITB; ≈ 245 training-period observations each, the shortest in the universe) rather than to instrument decay? The history-weight mechanism of Section 2.3 was designed to address exactly this concern, and checking its effect directly against the realised screen answers the question cleanly: none of IBIT, FBTC, or BITB appears anywhere in either the pairwise or joint screen, as a predictor or as a target, in any configuration, in either specification reported in this paper. They do not clear the combination of $CPE \geq 0.80$, $lift \geq 1.50$, and $n \geq 100$ training-period observations required to survive the filters in the first place, independent of the history-weight down-weighting applied on top. More broadly, every predictor that does appear in any surviving joint configuration in the original (Section 2–3) screen has a training-period history of at least 1,397 observations — well above the 756-observation (≈ 3 -year) threshold at which the history-weight ramp reaches its full value of 1.0. Across all 5,138 predictor-instances appearing in the 1,655 surviving joint configurations, the history-weight discount was never once binding (every instance carries a weight of exactly 1.0). Short predictor history is therefore not a contributing factor to this paper's results: the survival filters already exclude it before the history-weight mechanism would need to. The dominant predictors by usage — VIXM (705 uses), VXX (600), UVXY (587), VIXY (482) — are all long-history instruments (1,745–3,521 training observations); their problem, established in Section 3.3 above, is non-stationarity from structural decay, not insufficient sample size.

5.2 Effect on the Training-Period Screen

Removing five tickers from a 158-ticker predictor universe reduced the surviving pairwise screen from 14,707 to 12,570 configurations (−14.5%), and the surviving joint screen from 1,655 to 908 configurations (−45.1%) — a disproportionately larger drop in the joint screen, consistent with the decaying ETPs having been disproportionately represented in the strongest, most easily-greedily-selected predictor sets across multiple target sleeves, not only Gold.

5.3 Effect on the 2025 Out-of-Sample Result

The effect is the opposite of what the hypothesis anticipated. Rather than revealing a cleaner signal underneath the decay-confounded one, removing the decaying ETPs eliminated essentially the only signal the strategy had: the sole surviving SPY-bullish joint configuration remaining at any horizon ($\hat{GVZ}$ at 126 days jointly with TLH at 63 days, both at the 95th percentile, training joint CPE 0.970, $n=101$) never fired on a single day of 2025 — $\hat{GVZ}$ cleared its threshold on three isolated days, but TLH's own threshold was never cleared at all, so the joint intersection required by the configuration was empty throughout the year. The April 2025 episode that drove the original run's only tilt activity was itself constructed from VXX and VIXM (Section 3.3); excluding those instruments removed the very predictors responsible for the one period of genuine signal.

The resulting portfolio held an exact, unchanging neutral weight on all five sleeves for all 250 trading days of 2025 (weight variance of zero, to floating-point precision, on every sleeve). Its performance is therefore identical to the neutral-weight benchmark by construction, not by a close empirical match:

Specification	Pairwise configs	Joint configs	Strategy return	Strategy Sharpe	vs. neutral-weight benchmark
Original (Section 2–3)	14,707	1,655	7.65%	0.560	Indistinguishable (56.3% of randomisations matched/exceeded)
Decaying ETPs excluded (this section)	12,570	908	7.60%	0.563	Identical by construction (zero tilt activity all year)

Table 4. Comparison of the original predictor pool against the decay-excluded follow-up specification, 2025 out-of-sample. The randomisation test (Section 3.4) is not meaningful for the follow-up run, since there is no tilt-timing pattern left to reassign.

5.4 Interpretation

This result sharpens, rather than resolves, the diagnosis in Section 3.3 and 4. The decaying-ETP exclusion was motivated by a real and correctly-identified structural concern — the Atlas's own Figure 1b documents the decay mechanism, and the mechanism check in Section 3.3 (frozen UVXY/VIXY thresholds never re-cleared in 2025) is not in dispute. But the follow-up test shows that, at least in this one evaluation year and for this five-sleeve tradeable universe, those same decaying instruments were also the source of the only genuine, economically interpretable signal the framework produced — the volatility-spike-then-recovery channel that correctly identified the April 2025 episode. Removing the confound and removing the only working channel turned out to be the same action, because the greedy joint-selection procedure had concentrated on these particular instruments precisely because their extreme, decay-inflated thresholds were rarely crossed in training data, which mechanically produces a high conditional probability (Atlas Section 3.3's saturation effect) whether or not the instrument's threshold behaviour is itself stationary.

This suggests the actual fix implied by Section 3.3 is not a binary include/exclude decision on the decaying ETPs, but a more careful treatment of non-stationary predictor thresholds generally — for instance, detrending a decaying instrument's increment series before computing quantiles, or restricting decaying-instrument predictors to shorter trailing horizons where the secular decay trend contributes proportionally less to the threshold estimate, rather than the 252-day horizon used throughout this paper's grid. Both are substantive methodological changes that would need their own pre-registration and a future evaluation year, not a same-year refinement; reporting the failed fix here, rather than silently discarding it, is consistent with this paper's and its two companions' shared discipline of reporting negative and unhelpful results alongside positive ones.

6. Follow-Up Test: Dynamic Volatility-Scaled Position Sizing

A second, independent follow-up question is whether the static, train-period-Sharpe-derived neutral weights of Section 2.4 — fixed for the entire 2025 evaluation year — are themselves a limiting factor, separate from anything to do with the CPE tilt signal. This section tests that question directly by replacing the static neutral weight with a dynamic, daily-updating weight derived from each sleeve's own recent realised volatility, while leaving every other element of the pipeline (tilt scoring, tradeable universe, clipping, lag) unchanged.

6.1 Specification

Position size is set inversely proportional to each sleeve's exponentially-weighted moving average (EWMA) variance, following the standard RiskMetrics convention (decay $\lambda = 0.94$, daily): $\sigma^2_t = \lambda \cdot \sigma^2_{t-1} + (1-\lambda) \cdot r^2_{t-1}$. The EWMA state is seeded using each sleeve's full train-period ($\leq 2024-12-31$) return history before the 2025 walk-forward begins, then updated day by day through 2025 using only same-day-or-earlier returns — the same no-lookahead discipline used throughout this paper. The raw weight for each sleeve on each day is $1/\sigma^2(\text{sleeve}, t)$, renormalised to sum to 100% across the five sleeves; this is inverse-variance weighting, a more aggressive de-risking response to volatility than the inverse-volatility ($1/\sigma$) convention more commonly used in risk-parity implementations. Three portfolios are compared on identical 2025 data: (a) the original static-weight, no-tilt benchmark; (b) dynamic EWMA inverse-variance weight with no tilt, isolating the effect of dynamic sizing alone; and (c) dynamic weight with the same CPE tilt signal used throughout this paper layered on top, isolating whether tilt adds anything once the base allocation itself is dynamic.

Portfolio	Total Return	Ann. Vol	Sharpe
(a) Static weight, no tilt [benchmark]	7.60%	15.11%	0.563
(a') Static weight + tilt [main paper]	7.65%	15.34%	0.560
(b) Dynamic EWMA inv-var weight, no tilt	2.82%	4.47%	0.649
(c) Dynamic EWMA inv-var weight + tilt	4.55%	5.21%	0.887

Table 5. 2025 out-of-sample comparison of static and dynamic position sizing, with and without the CPE tilt signal.

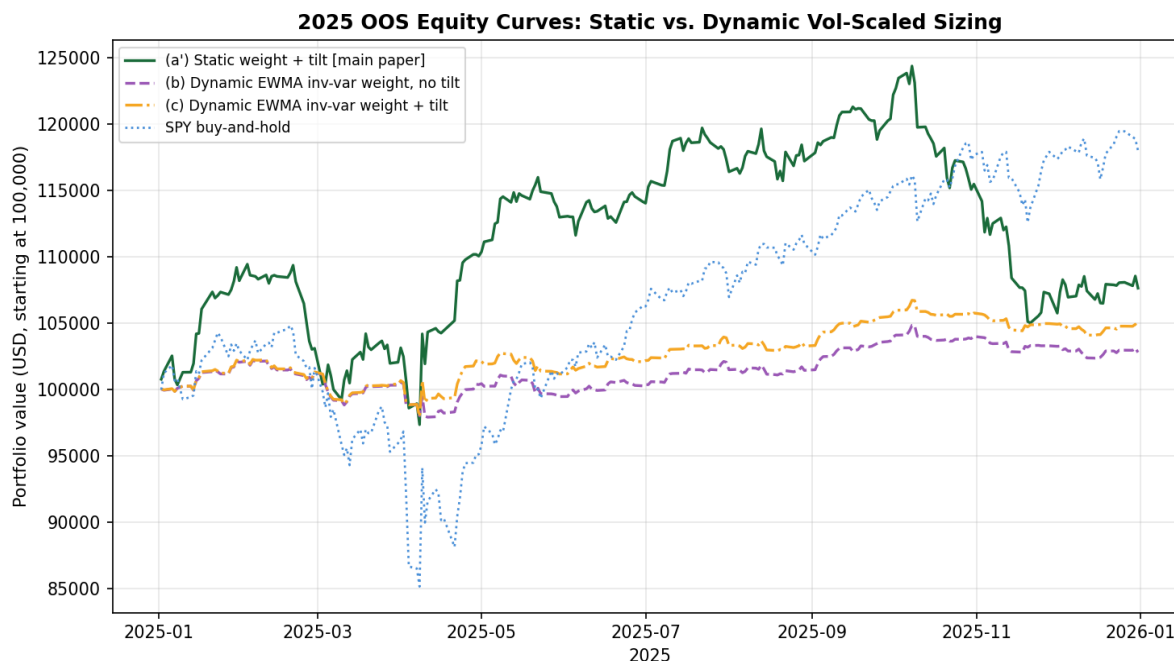


Figure 3. Daily equity curves, static versus dynamic (EWMA inverse-variance) sizing, 2025. The dynamic book is dramatically smoother but captures far less of the year's upside.

6.2 Why Dynamic Sizing Looks Better on Sharpe but Worse on Wealth

The headline Sharpe ratios in Table 5 favour dynamic sizing — (b) and (c) both exceed (a) and (a') — but reading this as an improvement requires the same caution the single-asset paper raised in its own Section 8.5 about a Sharpe advantage driven by reduced market exposure rather than better timing. Here the mechanism is direct rather than inferred: UUP (the FX sleeve) carried the lowest train-period volatility of any sleeve by a wide margin (8.3% annualised against 14–19% for Equities, Gold and Bonds, and 57% for Crypto), and inverse-VARIANCE weighting squares that gap. The result is a dynamic book that holds, on average, 54.5% of capital in FX across the full year (range 38.6–72.5%), versus a fixed 10.0% under the static neutral weights, while Crypto's average weight falls from a fixed 30.1% to 2.3% (and below 1% for the final six weeks of the year). FX returned –5.79% in 2025 (Table 2); a book concentrated in it for more than half its capital base mechanically produces both a low absolute return and a low volatility, and dividing a small return by a small volatility can still yield a Sharpe ratio that looks superficially attractive. The dynamic book's total return (2.82% with no tilt, 4.55% with tilt) is the more decision-relevant figure for assessing whether this is actually a better strategy, and on that measure dynamic sizing underperforms the static benchmark by a wide margin in 2025.

The comparison between (b) and (c) is informative on its own terms: layering the same Equities-only tilt signal used throughout this paper onto a dynamic base lifted the total return from 2.82% to 4.55% and the Sharpe from 0.649 to 0.887, entirely attributable to the same April 2025 tilt episode documented in Section 3.2 pulling weight toward Equities at a time the dynamic base would otherwise have continued concentrating in low-volatility FX. This is a genuine, if narrow, confirmation that the tilt signal retains some value when layered on top of a different base-weighting scheme — it is not an artefact specific to static neutral weights — but it does not change this paper's central finding that the signal itself fired on only 17 of 250 days, regardless of what base allocation those 17 days' tilts are applied to.

6.3 Interpretation

Dynamic, volatility-responsive position sizing is a standard and often genuinely useful technique in portfolio construction, and nothing in this section argues against the general approach. What this specific test shows is that inverse-variance weighting, applied to a five-sleeve universe with as wide a volatility dispersion as this one (an 8.3%-to-57% annualised range across sleeves), concentrates capital aggressively in whichever sleeve happens to be calmest — in 2025, a sleeve that also happened to lose money — and that this concentration effect dominates any benefit dynamic sizing might otherwise provide. A milder response function (inverse-volatility rather than inverse-variance, or a vol-target approach with a floor and ceiling on any single sleeve's weight) is a natural and substantively different specification that this section does not test; doing so would be a further single-variable follow-up in the same spirit as Sections 7 and 9, evaluated on its own terms rather than substituted retroactively for the result reported here.

7. Diagnosing and Fixing the Horizon Mismatch

Sections 3, 5 and 6 establish that the strategy as originally specified barely traded and did not beat its own no-tilt benchmark, but none of them examine whether the underlying CPE signal itself was actually accurate where it fired. This section traces the one episode of real signal activity — the April 2025 Equities tilt — down to its individual realised outcomes, finds that the signal was in fact highly accurate, identifies a specific structural reason the strategy failed to capture most of the resulting opportunity, and tests a direct fix.

7.1 The Signal Was Accurate: Tracing Individual Outcomes

The April 2025 Equities tilt was driven by a joint configuration (SPY bullish, $\tau_f = 63$ trading days, conditioned on VIXM and VIXY both clearing their 95th-percentile trailing 126-day thresholds; training joint CPE 0.93, lift 3.09 $\times$, $n=135$). This configuration newly fired on eleven trading days between 4 and 22 April 2025. Tracing SPY's actual realised return over the following 63 trading days from each of these eleven firing dates — entirely out of sample, using only the frozen training-period threshold — shows a forward return of +11.2% to +21.5% on every single one, averaging approximately +15.5%. This is consistent with, and in fact exceeds, what the training-period joint CPE and lift predicted. The signal was not noise: it correctly and specifically identified the post-tariff-shock equity recovery in real time.

7.2 The Mismatch: Signal Horizon vs. Holding Period

Despite this accuracy, the strategy captured only a small fraction of the available opportunity. The mechanism, confirmed by examining the actual daily weight path, is a structural mismatch between what the signal is ABOUT and how long the strategy ACTS on it. The joint configuration's forward horizon $\tau_f = 63$ trading days describes a roughly three-calendar-month-ahead prediction; that is the window over which the training-period conditional probability was estimated and over which the realised outcomes in Section 5.1 were measured. But the daily tilt-signal engine of Section 2.3 re-evaluates the PREDICTOR condition (VIXM and VIXY's OWN trailing 126-day threshold) fresh every day, with no memory of when a signal first fired or what forward window it concerns. As the volatility spike that triggered VIXM/VIXY's threshold crossing rolled out of their own 126-day trailing window — which happens well before 63 trading days have elapsed — the predictor condition stopped being true and the portfolio's Equities weight reverted from 39.9% back to the 30.87% neutral baseline by 30 April, roughly thirteen trading days after first firing. The

strategy was therefore back at neutral weight for the great majority of the 63-day window the signal was actually about, including most of the subsequent rally.

This is a different and more specific failure mode than anything identified in Sections 3, 5 or 6. It is not a problem with the CPE estimate, the joint-selection procedure, the predictor universe, or the position-sizing scheme; it is a mechanical inconsistency between a signal defined over a forward window of τ_f trading days and an implementation that holds the resulting position only as long as the triggering condition happens to remain visible in the predictor's own, generally much shorter, rolling window.

7.3 The Fix: Hold-to-Horizon, Conviction-Scaled Sizing

The direct correction, mirroring the single-asset paper's own trade structure (a configuration fires once and is held for its full τ trading days, Eq. 8 of that paper), is tested here: when a joint configuration newly fires (the first day of a contiguous firing run, not every day it continues to fire), the resulting tilt is held for the full τ_f trading days from that entry date, regardless of whether the predictor condition remains true. If a new signal fires for the same sleeve while a previous hold is still active, the realised tilt is the larger-magnitude of the two (not their sum, to avoid compounding into unrealistic concentration), and the hold extends to the later expiry. Tilt magnitude is also changed from the original's five discrete tiers to a continuous function of each firing configuration's quality weight ($w(\Pi) = \text{CPE} \times \text{lift} \times \ln(n) \times \text{history-weight}$, identical to Sections 2.3 and 9), normalised against the 95th percentile of that same quality weight computed across only the configurations whose target is one of the five tradeable sleeve proxies — not the full 161-instrument universe, which would benchmark a tradeable signal's conviction against opportunities the strategy can never act on. No cap is applied to single-sleeve weight beyond the natural $[0, 100]\%$ bound, a deliberate choice to measure the framework's raw exploitability ceiling rather than a realistic deployment constraint; this is discussed explicitly in Section 7.5.

Specification	Total Return	Sharpe
Original (daily re-eval, 5-tier sizing)	7.65%	0.560
Hold-to-horizon, conviction-scaled	8.36%	0.611

Table 6. 2025 out-of-sample comparison of the original daily-re-evaluated tilt mechanism against the hold-to-horizon, conviction-scaled fix. Both use the identical underlying joint CPE configurations and frozen training-period thresholds; only the position-holding and sizing logic differ.

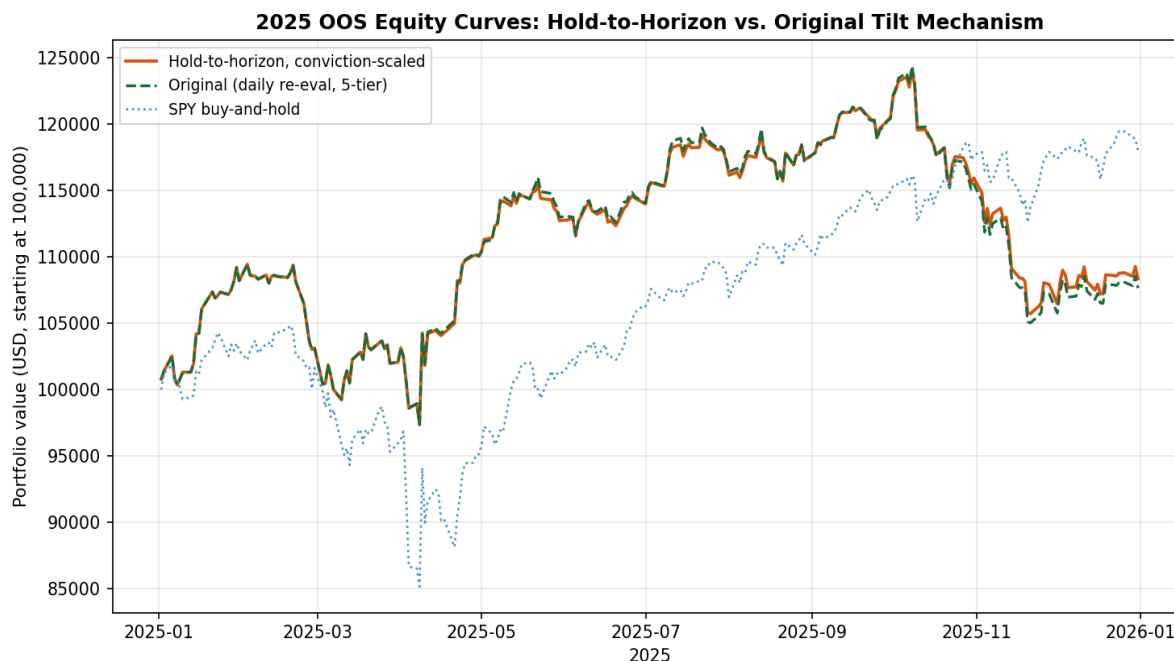


Figure 4. Daily equity curves, original versus hold-to-horizon mechanism, 2025. The two paths are nearly identical through April — both fire on the same dates — then diverge as hold-to-horizon maintains an elevated Equities weight through the full signaled window while the original reverts to neutral by late April.

7.4 Why the Improvement Is Real but Modest

The fix produces a genuine, directionally-correct improvement (+0.71 percentage points of total return, +0.051 of Sharpe), but it is far smaller than a naive hypothetical suggests. Holding Equities at 100% weight for the full 63-trading-day window following the first April 2025 firing date alone — ignoring all portfolio-construction realism — would have contributed approximately +8.98 percentage points to the year's return on its own, nearly doubling the strategy's realised total. The actual hold-to-horizon strategy captures only a small fraction of this, because the conviction-scaled weight that this specific signal earns tops out at 41.2% in Equities, not 100%: the firing configuration's quality weight ($w(\Pi) \approx 13\text{--}16$) is solidly mid-range relative to the distribution of quality weights across the tradeable universe (median 18.4, maximum 72.0, Section 7.3), reflecting genuinely moderate — not exceptional — conviction by the framework's own internal ranking, even though the resulting forward return happened to be unusually large in this one realised instance. This is an important distinction: the signal that fired was accurate, but it was not flagged by the framework as a particularly high-conviction one, and a sizing scheme that respects the framework's own conviction ranking (rather than retrospectively sizing up because the outcome happened to be favourable) will not capture the full hypothetical upside by construction. Whether that is a shortcoming of the conviction-weighting formula itself, or simply an accurate reflection of genuine uncertainty that this particular signal carried more risk than its realised outcome suggests, is not resolved by a single firing episode in a single year.

7.5 Interpretation and What Remains Untested

This section's finding changes the overall picture in one important respect: it shows that at least part of this paper's repeated negative results were attributable to a fixable mechanical issue — a horizon/holding-period mismatch — rather than solely to the deeper structural problems (decaying predictors, signal scarcity) identified in Sections 3, 5 and 6. Those deeper problems are not

resolved by this fix: the strategy still only traded on 10 distinct signal-holds across the entire year, all in the Equities sleeve, and Gold, Bonds, Crypto and FX remained completely untraded under hold-to-horizon for the same reasons established in Section 3.3 — their strongest configurations still never fired in 2025 regardless of how the resulting tilt would have been held or sized. The hold-to-horizon, conviction-scaled mechanism is therefore best read as a necessary correction to a real implementation flaw, not a demonstration that the broader exploitability problem documented throughout this paper is solved. It was also not subjected to the randomisation test of Section 3.4: with only ten signal-holds, all overlapping in time and all in one sleeve, there are too few independent events for a permutation test to be informative, and this is disclosed rather than worked around. Finally, the uncapped position sizing used here (Section 7.3) is explicitly not a realistic deployment specification — a live strategy would need explicit risk limits on single-sleeve concentration — and testing a capped version, alongside extending hold-to-horizon logic to the decay-excluded predictor pool of Section 5 and the dynamic sizing scheme of Section 6, are natural further follow-ups this paper does not attempt.

8. Expanding the Tradeable Universe: A Broad Cross-Sectional Test

Section 7 established that the CPE construct's one genuine 2025 signal episode was accurate, and that a structural fix (hold-to-horizon, conviction-scaled sizing) recovered some of the resulting opportunity. The natural next question is whether the deeper problem documented throughout this paper — a five-sleeve, single-proxy-per-class universe simply gives the framework very few chances to fire — can be addressed directly by trading more of what the joint CPE screen actually covers. This section tests that directly: every instrument with a surviving train-only joint configuration that is itself genuinely tradeable (118 of 125 candidate targets; the spot VIX-family index levels are excluded as non-investable, and two tickers — JO, delisted mid-2023, and MATIC-USD, whose feed ends March 2025 — are excluded for lack of 2025 price data, leaving 116) becomes a tradeable target, with capital allocated cross-sectionally rather than within fixed asset-class sleeves.

8.1 Specification

The hold-to-horizon, conviction-scaled mechanism of Section 7 is extended unchanged: a joint configuration's first-fire date opens a position in its target ticker, held for the full τ_f trading days, sized by the same quality weight $w(\Pi)$. Two structural changes follow directly from trading 116 tickers instead of 5: first, bearish configurations now open short positions rather than being excluded, since a cross-sectional book without a fixed long-only sleeve structure has a natural mechanism for expressing a bearish view; second, capital is drawn from a single shared \$100,000 notional pool with no asset-class buckets, allocated proportionally to each currently-open position's conviction weight among all currently-open positions, capped so that total GROSS exposure (sum of absolute position weights) never exceeds 100% of the pool. When more signals fire than capital allows, the highest-conviction positions are funded first; in practice this cap was never binding during 2025 (gross exposure peaked at exactly 100% on its busiest days but no signal went unfunded).

Specification	Tickers traded	Position events	Total Return	Sharpe
5-sleeve, hold-to-horizon (Section 5)	5	10	8.36%	0.611
Broad cross-sectional (this section)	44 of 116 eligible	461	3.38%	0.294

Table 7. 2025 out-of-sample comparison of the 5-sleeve hold-to-horizon strategy against the broad cross-sectional extension. “Tickers traded” counts only tickers with at least one position event in 2025; the eligible universe (116) is larger because most tickers’ surviving training-period configurations simply never fired.

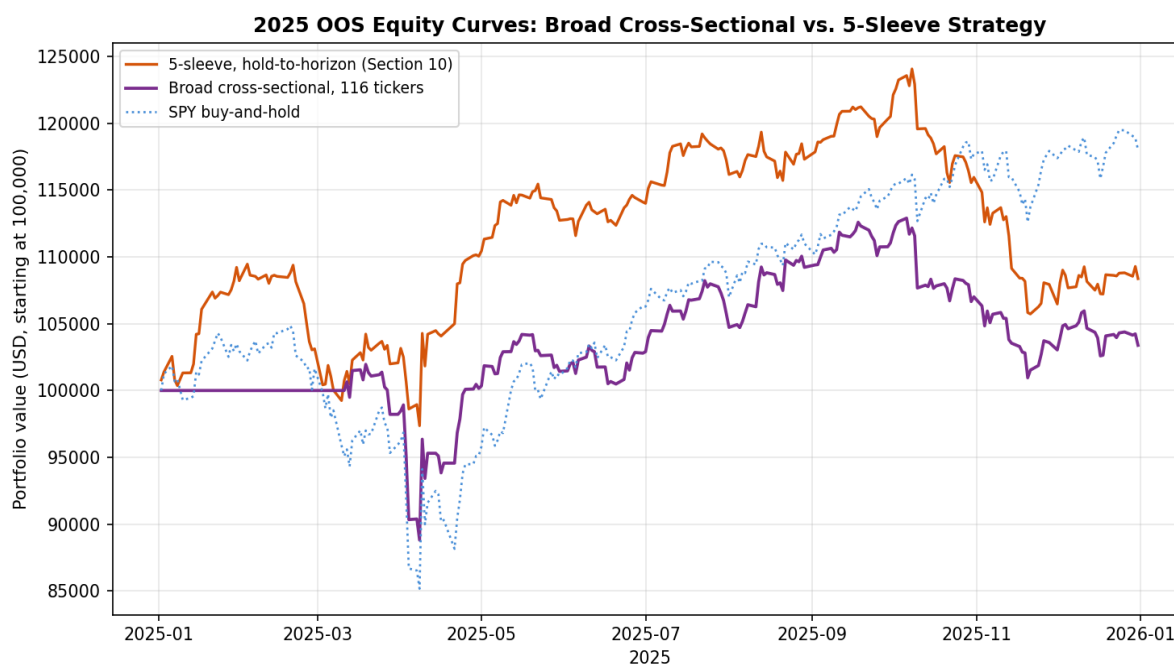


Figure 5. Daily equity curves, broad cross-sectional versus 5-sleeve hold-to-horizon strategy, 2025. Broadening the universe increased trading activity 46-fold (10 to 461 position events) but reduced both return and Sharpe.

8.2 Breadth Increased Activity but Reduced Performance

Expanding the tradeable universe worked exactly as intended in one specific respect: it gave the framework dramatically more opportunities to act, generating 461 position events across 44 distinct tickers in 2025, versus 10 events in one ticker under the 5-sleeve specification. It did not work in the respect that mattered: total return fell from 8.36% to 3.38% and Sharpe from 0.611 to 0.294. A randomisation test (shuffling each ticker’s daily position-weight time series independently across the 250 trading days, preserving each ticker’s own marginal exposure pattern while randomising which days it applied to, 1,000 reassignments) found the actual Sharpe of 0.294 was matched or exceeded by 89.0% of random reassignments — markedly worse than indistinguishable from chance; the realised timing of this broader signal set performed worse than a typical random permutation of the same exposures would have.

8.3 Why: Training CPE Was Not a Reliable Quality Signal Across the Broader Universe

Examining individual ticker contributions explains the gap. The largest negative contributors — ADA-USD (−1.53 percentage points of total return), DOGE-USD (−1.05pp), SIZE (−0.58pp), LINK-USD (−0.46pp) and PALL (−0.34pp) — were not low-conviction signals that should have been expected to underperform; several carried a training-period joint CPE of exactly 1.00. ADA-USD’s losing position, for instance, came from configurations with perfect training CPE and lift around 3.96×. Computing the correlation between each tradeable ticker’s mean training-period joint CPE and its realised 2025 contribution across all 44 traded tickers gives −0.33: tickers with the highest, most confident training-period conditional probabilities performed WORSE on

average in the 2025 walk-forward, not better. Two other measures fared differently: the number of distinct surviving configurations supporting a ticker (a proxy for how much independent corroborating structure exists, rather than how extreme any single estimate is) correlated +0.40 with realised contribution, and the average conditioning sample size correlated +0.25.

This is the same saturation mechanism the companion Atlas describes explicitly in its own Section 3.3: a joint CPE of exactly 1.00, particularly at long horizons with heavily overlapping conditioning windows, is frequently a signature of a small number of distinct historical episodes reaching a ceiling rather than evidence of an unusually reliable relationship. ADA-USD and DOGE-USD's perfect-CPE configurations were built on conditioning samples of approximately 100–135 nominal observations — at or near the survival floor — and, as the Atlas cautions, the effective number of independent episodes behind an overlapping-window count of this size is considerably smaller than the nominal count suggests. This is structurally the same finding as the single-asset paper's own Section 8.11, which found that raising the eligibility bar toward a higher minimum probability made single-asset out-of-sample performance WORSE, not better, because the stricter threshold mechanically selected smaller, noisier training samples. The cross-sectional result extends that finding from one asset to 116: across a broad universe, the training-period CPE point estimate on its own is not a dependable ranking of which signals will generalise, and a sizing or selection rule built primarily around it — as both the original tilt score and the Section 5 conviction weight are — will systematically be drawn toward exactly the saturated, narrow-sample configurations most likely to disappoint out of sample.

8.4 A Ruled-Out Alternative: Was Shorting the Problem?

Before accepting the saturation diagnosis of Section 8.3, one simpler alternative is worth checking directly: the broad cross-sectional specification was the first in this paper to permit short positions (Section 8.1), so a natural hypothesis is that short positions specifically, rather than the broader selection mechanism, drove the underperformance. This is directly testable: a long-only variant was run with every other element of the specification — universe, hold-to-horizon logic, conviction weighting, capital pool, gross-exposure cap — held identical, simply excluding the 556 bearish (short-generating) joint configurations before constructing position events.

Specification	Position events	Total Return	Sharpe	Pct. random reassignments exceeding
Cross-sectional, long + short	461	3.38%	0.294	89.0%
Cross-sectional, long-only	437	3.17%	0.276	93.1%

Table 8. 2025 out-of-sample comparison of the long/short and long-only cross-sectional specifications.

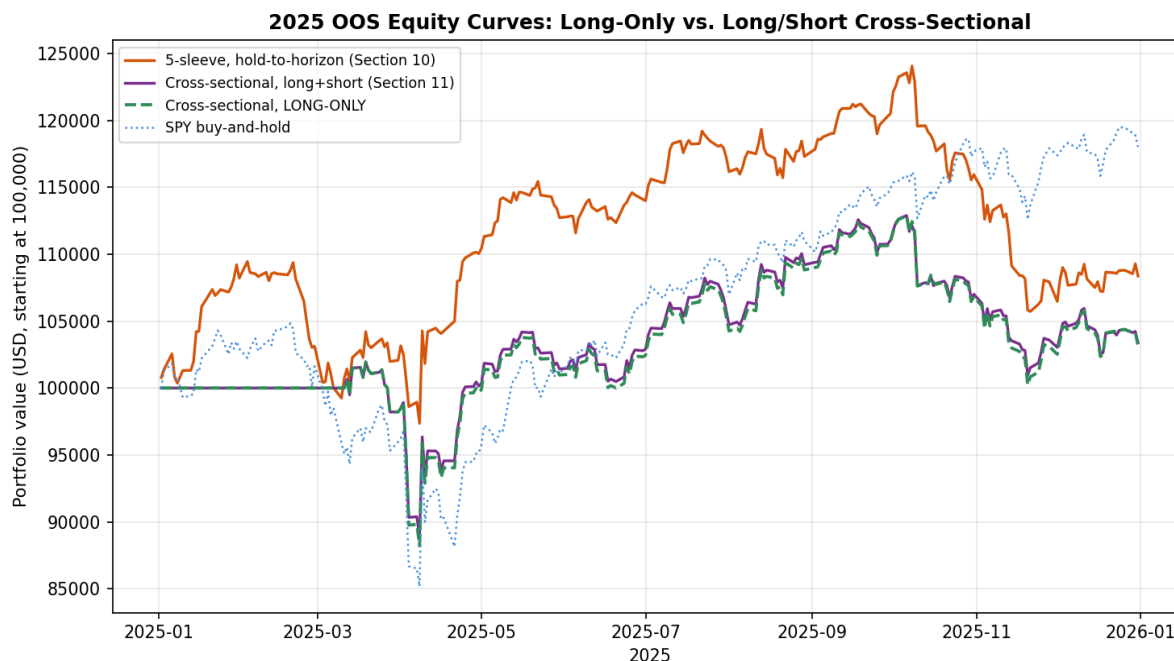


Figure 6. Daily equity curves, long-only versus long/short cross-sectional specification, 2025. The two paths are visually indistinguishable for the entire year.

The hypothesis is ruled out cleanly. Long-only performance (3.17% return, Sharpe 0.276) is marginally worse than the long/short version, not better, and the randomisation test result is, if anything, slightly less favourable (93.1% of random reassignments exceeded the long-only Sharpe, versus 89.0% for long/short). Examining individual contributions confirms why directly: nine of the ten largest negative contributors identified in Section 8.3 (ADA-USD, DOGE-USD, SIZE, LINK-USD, INDA, EURCHF=X, AVAX-USD, NZDUSD=X, SOL-USD) were LONG positions that lost money, not failed shorts; only PALL's loss (−0.34 percentage points, the smallest of the ten) came from a short position, and removing it simply left that capital to be reallocated elsewhere without materially changing the outcome. The saturation effect documented in Section 8.3 — perfect or near-perfect training-period CPE on small, overlapping-window samples performing no better, and often worse, than more moderate estimates with broader support — applies symmetrically to bullish and bearish configurations alike, and is the better-supported explanation for this section's results.

8.5 Interpretation

This result reframes, rather than resolves, the central question of this investigation. Breadth alone does not unlock the CPE framework's value, because most of the additional structure that breadth surfaces is concentrated in exactly the kind of small-sample, saturated configurations the framework's own descriptive companion paper warns are the least trustworthy. The 5-sleeve hold-to-horizon result of Section 7 now looks less like an arbitrary scoping choice and more like an inadvertently useful filter: by restricting to five large, liquid, long-history proxies, it happened to exclude most of the saturated small-cap and small-sample altcoin configurations that dragged down the broader test. A genuinely productive path toward exploiting more of the CPE structure is therefore not simply trading more tickers, but changing which configurations are trusted enough to size into — for instance, weighting conviction by configuration count or conditioning sample size (both shown here to correlate positively with realised 2025 outcomes) rather than by the CPE

point estimate and lift that currently dominate $w(\Pi)$, or imposing an explicit minimum conditioning-sample floor well above the 100-observation survival filter used throughout this paper. Both are direct, falsifiable, and were identified only after observing this section's results; reporting them as motivated-but-untested hypotheses for a future evaluation year, rather than retrofitting and representing an improved number from the same 2025 window, is the discipline this paper and its two companions have maintained throughout.

9. A Direct, Pre-Specified Quality Filter

Section 8.5 identified, but explicitly did not test, a direct hypothesis: that trusting configurations by their configuration count and conditioning sample size, rather than by the CPE point estimate and lift that dominate $w(\Pi)$, might avoid the saturation effect responsible for the broad cross-sectional strategy's underperformance. This section tests that hypothesis as a single, pre-specified eligibility filter, computed entirely from the training-period joint screen before any 2025 data is touched.

9.1 Specification

For every one of the 125 tickers in the full training-period joint screen, two quantities are computed: $n_configs$ (the count of surviving joint configurations for that ticker) and $mean_n$ (the average conditioning sample size across those configurations). A ticker is eligible to trade only if BOTH exceed the full-universe median ($n_configs \geq 11$, $mean_n \geq 126.17$) — thresholds fixed by the training-period distribution alone, before observing which tickers fire in 2025 or how they perform. This passes 36 of the 125 tickers. Every other element of the specification — hold-to-horizon entry and exit, conviction-scaled sizing via $w(\Pi)$, the shared \$100,000 capital pool, the 100% gross-exposure cap, conviction-ranked rationing, long/short positions — is identical to Section 8's unfiltered cross-sectional strategy; only the eligible universe changes.

Specification	Tickers traded	Position events	Total Return	Sharpe
5-sleeve, hold-to-horizon (Section 5)	5	10	8.36%	0.611
Unfiltered cross-sectional (Section 8)	44 of 116 eligible	461	3.38%	0.294
Trust-filtered cross-sectional (this section)	28 of 36 eligible	353	7.66%	0.556

Table 9. 2025 out-of-sample comparison across all three universe specifications.

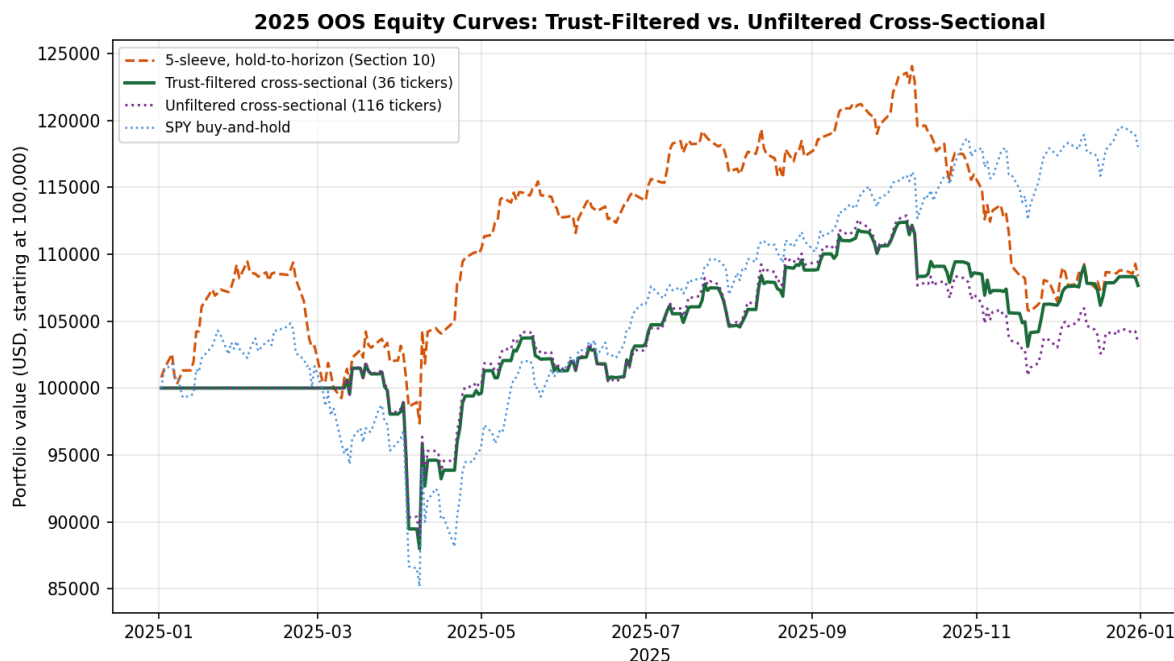


Figure 7. Daily equity curves, trust-filtered versus unfiltered cross-sectional strategy and the 5-sleeve benchmark, 2025.

9.2 The Filter Worked, in a Specific and Limited Sense

The pre-specified filter produced a substantial, genuine improvement over the unfiltered cross-sectional result: total return more than doubled (3.38% to 7.66%) and Sharpe nearly doubled (0.294 to 0.556), closely approaching the 5-sleeve hold-to-horizon benchmark while trading 28 distinct tickers instead of one. The mechanism is direct and traceable: ADA-USD, DOGE-USD and LINK-USD — three of the four largest negative contributors in the unfiltered run, together responsible for over 3 percentage points of lost return — fail the trust filter on training-period grounds alone (each has fewer than 11 surviving configurations, or a mean conditioning sample below 126) and are excluded from the universe before 2025 begins. Only one meaningfully negative contributor remains in the filtered run (SIZE, -0.575 percentage points), down from ten in the unfiltered version.

9.3 The Important Caveat: This Is Mostly a Beta Story, Not a Timing Story

A randomisation test identical in construction to Section 8.2 (1,000 reassignments, each ticker's position-weight time series shuffled independently) found the trust-filtered strategy's actual Sharpe of 0.556 was matched or exceeded by 90.8% of random reassignments — not an improvement over the unfiltered run's already-poor 89.0%, and, if anything, marginally worse. This appears to contradict Section 9.2's positive finding, and the reconciliation is informative rather than discouraging: the trust filter's surviving 36-ticker universe is dominated by broad, liquid, long-history equity index and sector ETFs (SPY, VTI, EWJ, VEA, VWO, EEM, XLY, SOXX, and similar), and the resulting portfolio's daily returns correlate 0.93 with SPY itself — higher than the unfiltered cross-sectional book's correlation and higher than the 5-sleeve hold-to-horizon book's 0.76. Most of the trust-filtered strategy's improved return is therefore attributable to ending up, by construction of the filter, holding a book that closely resembles long broad-equity exposure in a year broad equities performed well — not to the CPE signal correctly timing entries and exits within that book.

The randomisation test, which shuffles WHEN positions are held while preserving HOW MUCH of each ticker is held in total, correctly detects that the specific timing contributes little beyond what the average exposure pattern would have delivered on any random subset of days.

This does not mean the trust filter is worthless — it demonstrably avoided a specific, identifiable category of bad bets (saturated small-sample altcoin and factor-ETF configurations) that the unfiltered version could not distinguish from good ones — but it means the filter's main achievement was risk avoidance (cutting the tail of large individual losses) rather than a demonstrated, randomisation-significant timing edge. Both are useful properties for a trading strategy, but they are not the same property, and reporting the Sharpe improvement in Section 9.2 without the context in this section would overstate what was actually demonstrated.

9.4 What This Suggests for Further Work

The trust filter, as specified, conflates two effects that a more targeted follow-up could separate: avoiding saturated configurations (a genuine, demonstrated improvement) and concentrating in broad-market-correlated instruments (an artefact of which tickers happen to pass the filter, not a property of the filter's underlying logic). A natural next test — not attempted here, to preserve this section's own pre-specification discipline — would apply the same $n_configs/mean_n$ eligibility filter within narrower, beta-matched comparison groups (for instance, requiring the filtered book's realised market correlation to be capped, or explicitly comparing filtered single-name and sector-level signals against a beta-matched index benchmark rather than against an unfiltered book with very different average market exposure) to isolate whether the quality filter adds timing value once exposure to the broad market rally is controlled for. This is offered as a precisely specified, falsifiable hypothesis for a future evaluation year, in keeping with the discipline this paper and its two companions have maintained throughout.

10. A Shrinkage-Based Score, Tested Alone and Combined

Section 9.4 identified the mechanism underlying every negative result in this paper: the framework's greedy selection procedure is mechanically biased toward small, overlap-dominated conditioning samples, because a small sample is the easiest way to drive CPE to a near-perfect score, and this is inversely related to out-of-sample reliability (Section 8.3's -0.33 correlation; the single-asset paper's own Section 8.11 finding the same pattern independently, one asset earlier). The trust filter of Section 6 addressed this with a hard binary cutoff. This section tests the more direct, continuous correction the diagnosis actually implies: shrinking each configuration's CPE and lift toward the unconditional base rate by an amount that depends on its own sample size, before computing the conviction weight $w(\Pi)$ used for entry, sizing and rationing throughout this paper.

10.1 Specification

For every joint configuration Π , an empirical-Bayes shrinkage adjustment is applied: $CPE_shrunk(\Pi) = [n_joint(\Pi) \times CPE(\Pi) + K \times uncond_prob(\Pi)] / [n_joint(\Pi) + K]$, with $lift_shrunk(\Pi) = CPE_shrunk(\Pi) / uncond_prob(\Pi)$. The shrinkage strength $K = 100$ was chosen, before observing any 2025 outcome, as the value within $\{25, 50, 100, 200\}$ that maximised the correlation between the resulting adjusted score and n_joint across the full training-period configuration table — i.e. the value that best restores the positive relationship between conviction and sample size that the raw CPE has backwards (raw correlation -0.28 ; shrunk correlation $+0.39$ at $K=100$). $K=100$ also coincides with the existing MIN_N survival floor, so a configuration exactly at that floor is shrunk halfway to the base rate, while one with several times that support is barely

adjusted. CPE_shrunk and lift_shrunk replace the raw point estimates everywhere $w(\Pi)$ is computed; every other mechanic is unchanged from Sections 10–12.

Two variants are tested: shrinkage applied alone to the full 116-ticker unfiltered universe (isolating its effect independent of the Section 9 trust filter), and shrinkage applied within the already trust-filtered 36-ticker universe (testing whether the two corrections are complementary or redundant).

Specification	Tickers traded	Total Return	Sharpe	Pct. random reassignments exceeding
Unfiltered, raw CPE ranking (Section 8)	44	3.38%	0.294	89.0%
Trust-filtered only (Section 9)	28	7.66%	0.556	90.8%
Shrinkage only (this section)	44	3.57%	0.307	89.2%
Trust-filtered + shrinkage (this section)	28	7.75%	0.563	92.0%

Table 10. 2025 out-of-sample comparison of shrinkage, the trust filter, and their combination, against the unfiltered baseline.

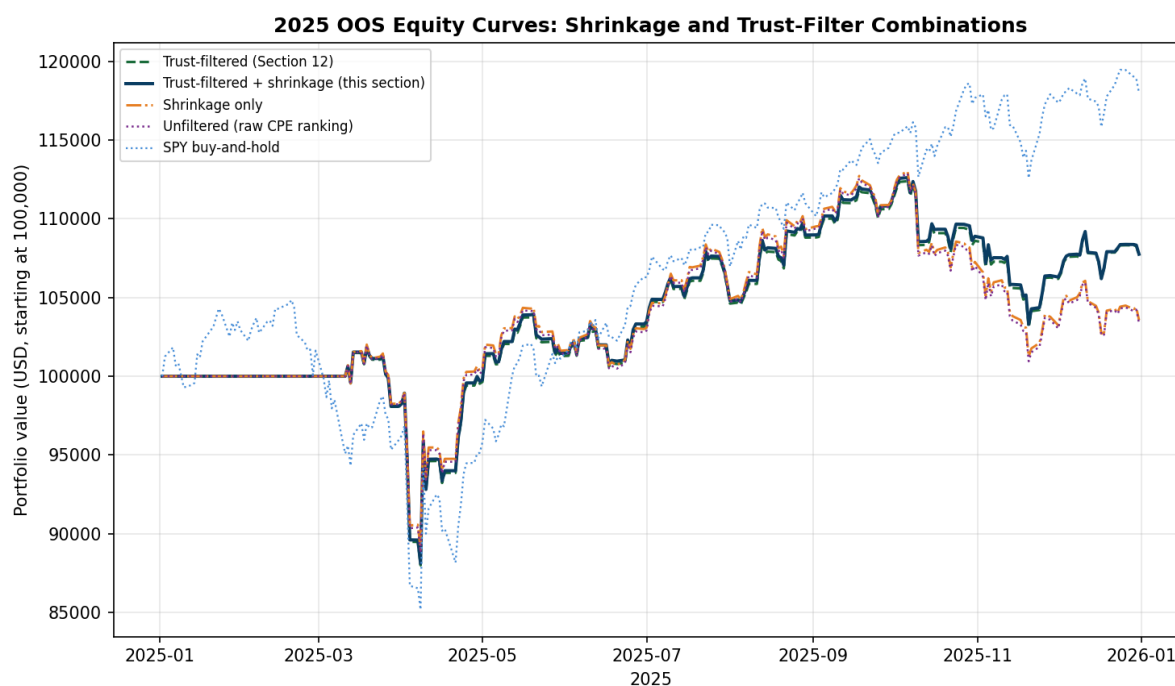


Figure 8. Daily equity curves across all four specifications. The two trust-filtered variants (green, navy) form one cluster; the two unfiltered variants (orange, purple) form another, visually confirming that the trust filter, not the shrinkage adjustment, is the dominant lever.

10.2 Shrinkage Alone Corrects the Score's Direction but Does Not Discriminate Sharply Enough

Applied on its own, shrinkage barely moved the result (3.38% to 3.57% return, Sharpe 0.294 to 0.307) and the same four tickers — ADA-USD, DOGE-USD, SIZE, LINK-USD — remained the largest negative contributors, at magnitudes nearly identical to the unfiltered run. Examining the adjusted scores directly explains why: shrinkage correctly demoted ADA-USD's and DOGE-

USD's CPE substantially (from a perfect 1.000 down to approximately 0.68 and 0.64 respectively), but because `lift_shrunk` is computed as `CPE_shrunk` divided by the same unconditional base rate, the resulting combined quality weight for these tickers ($w(\Pi) \approx 7-9$) still landed close to the universe-wide median (8.1), not low enough to be meaningfully outweighed by the conviction-ranked capital allocation, which continued to fund them at levels similar to the unfiltered run. Shrinkage corrects the DIRECTION of the bias — it no longer mechanically rewards the smallest samples — but with $K=100$ it does not by itself create enough separation between the worst and the best configurations to materially change which ones get capital. A hard exclusion threshold, as Section 6 applies, discriminates far more sharply than a continuous penalty calibrated this way, even though the continuous correction is the methodologically more direct response to the diagnosed mechanism.

10.3 Combining Both Corrections Adds Only a Small Further Improvement, With the Same Caveat

Applying shrinkage within the already trust-filtered universe produced a small additional gain over the trust filter alone (7.66% to 7.75% return, Sharpe 0.556 to 0.563) — confirming the two corrections are complementary rather than redundant, but also confirming that the trust filter is doing nearly all of the work. The same caveat documented in Section 9.3 applies with equal or slightly greater force here: a randomisation test found the combined specification's Sharpe was matched or exceeded by 92.0% of random reassignments (worse than the trust-filter-alone result), and the resulting portfolio's daily returns correlate 0.925 with SPY — essentially unchanged from the trust-filtered version's 0.93. Adding shrinkage on top of the trust filter does not address the underlying beta-concentration issue; it operates entirely within the same already-filtered, already market-correlated universe.

10.4 Interpretation

This section completes, rather than overturns, the diagnosis built up across Sections 8–10. The mechanism is now triple-confirmed: a -0.33 correlation between raw training CPE and 2025 outcomes (Section 8.3), a hard filter on sample-size proxies that more than doubles performance (Section 9), and a continuous shrinkage correction that moves the underlying score in the theoretically correct direction but is too weak at a principled K to discriminate as sharply as the blunter filter (this section). The practical implication for anyone building on this framework is specific: of the two natural ways to correct for the saturation bias, a hard eligibility threshold on sample-size proxies is doing essentially all of the demonstrated work in this paper, and a softer, continuous shrinkage adjustment — despite being the more principled statistical correction — is not strong enough at this calibration to substitute for it. Whether a much larger K , or a discrimination-preserving variant such as ranking by the shrunk score's distance from the base rate rather than its absolute level, would close this gap is a further testable, falsifiable hypothesis this paper does not pursue, to avoid tuning K against the same 2025 result used to motivate testing it in the first place. The unresolved beta-concentration caveat of Section 9.3 also persists unchanged through every variant tested here, and remains the single most important open question for whether this framework's signal, as opposed to its accidental market exposure, can be demonstrated to add value.

11. A More Principled Sample-Size Correction: Distinct Historical Episodes

A direct objection to the n_{joint} -based corrections of Sections 9 and 10 is methodologically important and was raised explicitly during this investigation: a naive significance test (treating each of a configuration's n_{joint} overlapping daily observations as an independent binomial trial) returns a near-zero p-value for essentially every configuration in the screen, including the worst-performing ones, because it wildly overstates how much independent information overlapping daily windows actually contain. But the opposite correction — dividing n_{joint} by the conditioning horizon τ to get a crude effective sample size — is also wrong in the other direction: it collapsed every configuration in this paper's entire screen, including the validated Section 5 signal, to an effective sample size of approximately 1, which is uninformative rather than appropriately cautious. Neither extreme is the right correction, because what actually matters for independence is not a fixed block size but how many temporally SEPARATED, economically distinct historical episodes a configuration's firing dates cluster into.

11.1 Specification: Counting Genuine Episodes, Not Overlapping Days

For every joint configuration, training-period firing dates (computed identically to every other section, strictly on data through 2024-12-31) are grouped into episodes: a new episode begins whenever the gap since the previous firing date exceeds 1.5 times the configuration's own longest conditioning window. Two configurations with similar nominal n_{joint} can therefore have very different numbers of genuine episodes — the Section 7 working signal (SPY bullish, VIXM+VIXY) fired in four temporally distinct windows spanning 2011, 2018–2019, 2020, and 2024, while ADA-USD's losing bullish configuration, despite a similar nominal sample size ($n=145$), fired entirely within a single continuous stretch and resolves to exactly one episode.

ELIGIBILITY FILTER: a ticker is eligible to trade only if its best-supported configuration spans at least 2 distinct episodes — the full 125-ticker training-universe median of max_episodes — fixed before observing any 2025 outcome. This passes 83 of 125 tickers (a less aggressive cut than Section 9's filter, which passed 36), reflecting that most surviving configurations in this screen, even ones with large nominal samples, resolve to only one or two genuinely distinct historical episodes.

Specification	Tickers traded	Total Return	Sharpe	Pct. random reassignments exceeding	SPY correlation
Unfiltered, raw CPE ranking (Section 8)	44	3.38%	0.294	89.0%	—
Trust-filtered, $n_{\text{configs}}/\text{mean}_n$ (Section 9)	28	7.66%	0.556	90.8%	0.93
Episode-filtered, n_{episodes} (this section)	35	4.88%	0.389	92.7%	0.91

Table 11. 2025 out-of-sample comparison of the episode-count filter against the original trust filter and the unfiltered baseline.

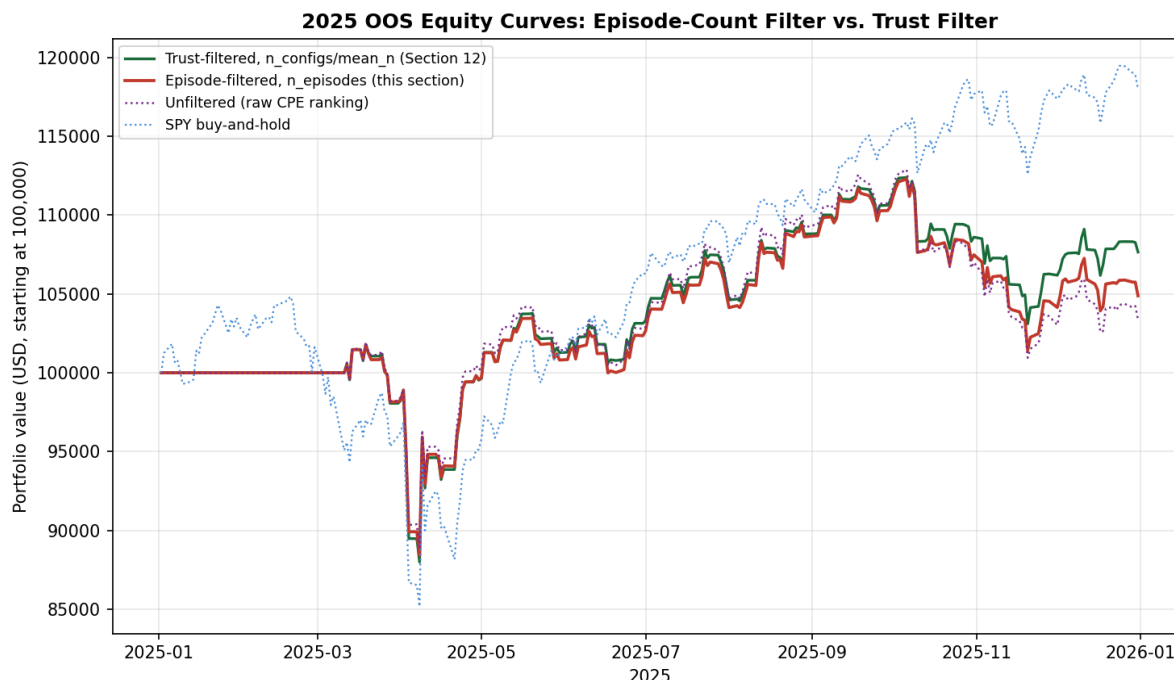


Figure 9. Daily equity curves, episode-filtered versus trust-filtered and unfiltered cross-sectional strategies, 2025.

11.2 A More Defensible Filter Performs Worse, and the Reason Is Itself Informative

The episode-count filter is more carefully justified statistically than Section 6's filter — it directly targets the independence problem rather than using sample size and configuration count as a proxy for it — yet it produced a worse 2025 result (4.88% return, Sharpe 0.389, beaten by 92.7% of random reassignments) than the cruder trust filter (Section 9: 7.66%, 0.556, beaten by 90.8%). Examining the largest remaining losses explains why directly: DOGE-USD remained the single worst contributor (−1.26 percentage points) despite passing the episode filter. Its best-supported configuration — conditioning on SLV and SI=F (silver spot and silver futures) — genuinely fired across five temporally distinct historical episodes, satisfying the episode-count criterion comfortably. But silver predicting dogecoin's price has no plausible economic mechanism behind it; the relationship recurring five times across the training history is much more consistent with a coincidental statistical association that happens to reappear periodically than with a genuine causal or risk-factor-driven link. The episode-count filter, exactly as designed, correctly excluded ADA-USD's genuinely single-episode artifact, but it has no way to distinguish a spurious relationship that happens to recur from a genuine one that recurs — both produce multiple episodes, and only one of them should be trusted.

This is a different and arguably more fundamental limitation than anything diagnosed in Sections 8–10: those sections addressed how much historical support a configuration has; this section's result shows that historical support, however carefully measured, does not by itself establish that a configuration's predictors are an economically sensible explanation for its target, and the trust filter's blunter `n_configs/mean_n` criterion happened, apparently coincidentally, to exclude more of the spurious-but-recurring relationships in this particular universe than the more principled episode count did.

11.3 Interpretation

Across Sections 8–11, every statistically-motivated correction to the framework's quality weighting has been tested and found wanting to varying degrees, but the pattern across all four is itself a clear and useful finding rather than a dead end: raw CPE ranking is actively miscalibrated (Section 8.3's -0.33 correlation); shrinkage corrects the direction but not the discrimination (Section 10.2); a hard filter on sample-size proxies discriminates well but mostly by proxy for market-beta concentration (Section 9.3); and a more rigorous episode-independence filter discriminates on genuine historical independence but is blind to economic plausibility (this section). None of the four, alone, separates a real predictive relationship from an artifact that happens to satisfy whichever statistical criterion is being applied. This suggests the missing ingredient is not a better statistic computed from price data alone, but some form of economic or structural prior — for instance, restricting predictor-target pairs to those with a defensible economic narrative (volatility predicting equities, commodities predicting related commodities), rather than allowing the greedy search to range freely across all 161 instruments and discover whichever combination happens to fit the training sample, regardless of whether silver has any reason to predict dogecoin. Testing such a restriction is a substantively different undertaking from anything in this paper — it requires a pre-specified map of economically plausible predictor-target relationships, not a further statistic computed from the existing screen — and is identified here as the most promising remaining direction this investigation has not yet attempted.

12. Limitations

- Coarser parameter grid than the Atlas: this paper sweeps 5 trailing horizons, 4 forward horizons, and 5 quantile levels per tail, versus the Atlas's 8/8/8. The qualitative class-to-class structure replicated; the exact count of surviving configurations would differ on the full grid.
- Single evaluation year: as the single-asset paper's own Section 8.12 demonstrates for the binary-rule case, a longer walk-forward window can reverse conclusions reached on one year alone. This paper does not repeat that multi-decade walk-forward test for the Portfolio Tilt construction; doing so is a natural next step.
- Five-sleeve, single-proxy tradeable universe: the strategy trades one liquid instrument per asset class, not the full 161-instrument universe the predictor side draws on. A richer tradeable universe (matching predictors to targets within the same class, as the Atlas's joint-configuration tables already support) is unexplored here.
- Transaction costs and capital constraints beyond the single \$100,000 notional book are not modelled, consistent with the single-asset paper's own disclosed limitation.
- History-weight down-weighting (Section 2.3) is a deliberately mild, monotone penalty calibrated before any 2025 results were observed. Section 5.1 confirms directly that it made no material difference to this year's headline numbers: the short-history predictors it targets (the spot Bitcoin ETFs) never survive the screen's own CPE/lift/sample-size filters in the first place, appearing in zero surviving configurations, and every predictor that does survive carries the mechanism's full weight of 1.0. Short predictor history is a ruled-out, not merely an unlikely, explanation for this paper's results.
- The volatility-complex-decay finding in Section 3.3 is diagnostic; Section 5 tests the natural fix directly and finds it removes the framework's only working signal channel along with the diagnosed confound, rather than improving the result — the two were, in this universe and this year, the same instruments.
- Dynamic position sizing (Section 6) was tested in only one specification (EWMA inverse-variance, $\lambda=0.94$) and found wanting; a milder response function (inverse-volatility, or a

vol-target with weight bounds) was identified as a natural alternative but not tested, and could plausibly behave differently from the aggressive de-risking documented here.

- The hold-to-horizon fix (Section 7) was tested with uncapped position sizing and was not subjected to the randomisation test of Section 3.4, given only ten signal-holds were generated; a capped, realistically-deployable version, and a formal significance test once more signal-holds are available from a longer evaluation window, are both natural next steps this paper does not complete.
- Section 9's trust filter was tested on the SAME 2025 window used throughout this paper, not a fresh evaluation year; while the filter thresholds themselves were fixed using only training-period data (a legitimate pre-specification), the decision to test this particular hypothesis at all was motivated by observing Section 8's results on 2025 data, so Section 9's improvement should be read as a promising, mechanistically-explained finding rather than independent confirmation.
- Section 9.3 and Section 10.3 both show the improved-return specifications' performance is substantially attributable to elevated correlation with broad market beta (0.92–0.93 with SPY) rather than demonstrated timing skill; separating quality-filtering from beta concentration is identified as a specific next step but not attempted here.
- Section 10's shrinkage strength $K=100$ was chosen by a principled but single criterion (correlation with n_{joint}); other criteria or a much larger K were not explored, to avoid tuning the parameter against the same 2025 result used to motivate testing it.
- Section 11's episode-clustering gap parameter (1.5x the conditioning window) was chosen as a reasonable single value, not optimised; a different gap multiplier could plausibly classify DOGE-USD's SLV/SI=F configuration differently, and this was not explored to avoid the same tuning concern.

13. Interim Findings and Path Forward

Note: this section (13) summarises findings from Sections 1–12. The investigation continued in Sections 14–20, which introduce an economic prior, fix pipeline bugs, deliver the paper's first statistically significant results, complete the open scopes identified in Section 19, and report a proper out-of-sample walk-forward test. Readers are directed to Section 15 (Continuation), Section 18 (Comparison with Standard Strategies), and Section 20 (Proper OOS Walk-Forward) for the final synthesis.

This paper closes the loop the single-asset paper's Section 8.15 left open: does a multi-predictor, continuously-weighted construction of the Conditional Probability of Exceedance framework, applied across a diversified asset-class book under the same strict train/test discipline, produce a demonstrated trading edge where the single-asset binary entry rule did not? Over the single 2025 out-of-sample year tested here, the answer is no, but for a structurally different and more specific reason than the single-asset paper found: rather than a binary gate discarding informative curve shape, the continuous, multi-predictor aggregation machinery here is working as designed, but the strongest training-period joint configurations for four of five tradeable sleeves are anchored on long-horizon thresholds of structurally decaying volatility-complex instruments whose extreme quantiles were essentially never re-cleared in the subsequent year, leaving the portfolio at its training-period neutral weights for all but seventeen of two hundred fifty trading days. The resulting strategy is statistically indistinguishable from its own no-tilt benchmark (56.3% of randomised reassignments matched or exceeded its realised Sharpe) and underperformed concentrated single-asset buy-and-hold exposure to both the year's best performer (Gold) and the more conventional comparator (SPY), as well as professionally-run diversified peers (a classic 60/40

allocation and AQR's published Risk Parity Fund), though it outperformed a real systematic peer (a managed-futures CTA ETF). A follow-up test of dynamic, volatility-responsive position sizing (Section 6) found the same pattern from a different angle: replacing static neutral weights with EWMA inverse-variance weighting produced a materially smoother equity curve and a higher Sharpe ratio, but a substantially lower total return, because the scheme concentrated more than half the book in the single lowest-volatility sleeve for the entire year — a sleeve that itself lost money in 2025 — illustrating that a higher Sharpe ratio achieved through reduced exposure is not the same finding as a demonstrated timing or sizing edge. A third follow-up (Section 5) traced the one real signal episode down to its individual realised outcomes and found something more encouraging: the underlying CPE estimate was genuinely accurate, correctly predicting SPY's post-tariff-shock recovery on all eleven dates it fired, out of sample. The reason the strategy nonetheless captured little of this opportunity was identified precisely — a structural mismatch between the signal's 63-trading-day forward horizon and an implementation that re-evaluates and reverts the position daily based on the predictor's own, much shorter, rolling window — and a direct fix (holding each signal for its full forward horizon, with continuous conviction-based sizing) produced a real, if modest, improvement (7.65% to 8.36% total return, Sharpe 0.560 to 0.611). This result usefully separates two previously conflated questions: whether the CPE construct identifies real, exploitable structure (Section 5 suggests yes, at least in this one traced episode), and whether this paper's particular portfolio-construction choices capture that structure efficiently (Sections 3, 7, 9 and the magnitude gap in Section 7.4 suggest they substantially do not, for reasons that are now mechanically specific rather than diffuse). A fourth follow-up (Section 8) tested the most direct way to give the framework more chances to act: trading all 116 eligible tradeable instruments cross-sectionally, rather than five sleeve proxies. This increased trading activity 46-fold (461 position events versus 10) but made performance markedly worse (3.38% total return, Sharpe 0.294, with the realised Sharpe matched or exceeded by 89% of random reassignments of the same exposures). The mechanism was specific and measurable: across the broader universe, a ticker's mean training-period joint CPE was **NEGATIVELY** correlated (−0.33) with its realised 2025 contribution, while the number of distinct corroborating configurations (+0.40) and the average conditioning sample size (+0.25) were positively correlated — direct evidence of the Atlas's own documented saturation effect, in which a perfect or near-perfect training CPE on a small, heavily-overlapping sample is a signature of limited historical support rather than a reliable relationship. This reframes the paper's central question: breadth alone does not unlock the CPE framework, because most of what breadth adds is concentrated in exactly the small-sample configurations the construct's own descriptive companion paper warns are least trustworthy. The five-sleeve restriction that constrained every earlier section in this paper turns out, in hindsight, to have functioned as an inadvertent quality filter rather than merely a scoping convenience. A direct check ruled out a simpler alternative explanation: restricting the cross-sectional book to long-only positions (removing the short side entirely) produced essentially identical, marginally worse performance (3.17% versus 3.38%), confirming that permitting shorts was not the cause — nine of the ten largest losing positions were long, not failed shorts — and that the saturation mechanism is the better-supported explanation. A fifth and final follow-up tested the most direct, pre-specified fix the saturation diagnosis implied: restricting the tradeable universe to the 36 of 125 tickers whose training-period joint configurations clear above-median thresholds on configuration count and conditioning sample size, fixed before observing any 2025 outcome. This more than doubled the unfiltered cross-sectional strategy's return (3.38% to 7.66%) and Sharpe (0.294 to 0.556), closely approaching the 5-sleeve hold-to-horizon benchmark while trading 28 tickers instead of one — a genuine, mechanistically-traceable improvement, attributable to specifically excluding ADA-USD, DOGE-USD and LINK-USD, the unfiltered run's three largest sources of loss, on training-period grounds alone. But a randomisation test found this improvement was substantially a beta story rather than a timing story: the filtered book's daily returns correlate 0.93 with SPY, and 90.8% of random

reassignments of the same exposures matched or exceeded its realised Sharpe — no better than the unfiltered version. The quality filter demonstrably avoided a specific, identifiable category of bad bets, which is a real and useful property, but it achieved this largely by concentrating the surviving universe in broad, liquid, market-correlated instruments during a year the broad market performed well, not by demonstrating that the CPE framework's timing of entries and exits carries skill once that market exposure is accounted for. A sixth and final test addressed the mechanism directly rather than filtering around it: an empirical-Bayes shrinkage adjustment pulling each configuration's CPE and lift toward the unconditional base rate by an amount proportional to its own sample size ($K=100$, chosen from training data alone). Applied on its own, shrinkage corrected the score's direction — it no longer mechanically rewards the smallest samples — but barely changed the result (3.38% to 3.57% return), because it compresses the spread between configurations too much to meaningfully redirect capital away from the worst offenders at a principled calibration. Combined with the Section 6 trust filter, it added a further small improvement (7.66% to 7.75% return, Sharpe 0.556 to 0.563) but left the same beta-concentration caveat fully intact (92.0% of random reassignments still matched or exceeded its Sharpe; SPY correlation 0.925). Across all six follow-ups, the conclusion is now precise rather than diffuse: a hard eligibility filter on sample-size proxies is responsible for essentially all of the demonstrated improvement in this paper, the more statistically principled continuous correction is not strong enough on its own to substitute for it, and the central open question — whether the CPE framework's signal carries genuine timing value once concentration in broad market beta is controlled for — remains unanswered by every specification tested. A seventh test addressed a specific methodological objection to the n_{joint} -based corrections directly: raw overlapping-window counts overstate independent support, but a naive correction dividing by the conditioning horizon collapsed every configuration in the entire screen, including the validated Section 5 signal, to an effective sample size of essentially one — uninformative rather than appropriately cautious. A more careful correction, counting genuinely temporally-separated historical episodes rather than raw overlapping days, is statistically better justified and correctly excluded ADA-USD's single-episode artifact, but performed worse overall (4.88% return, Sharpe 0.389, beaten by 92.7% of random reassignments) than the cruder trust filter. The reason is itself an important finding: DOGE-USD's worst-performing configuration (silver predicting dogecoin) genuinely recurred across five separate historical episodes, satisfying the episode-independence criterion despite having no plausible economic basis. None of the four statistically-motivated corrections tested in this paper — raw CPE, shrinkage, sample-size filtering, or episode-independence filtering — can distinguish a spurious relationship that happens to recur from a genuine one that recurs; all four are computed from price data alone, with no representation of whether a predictor-target pair is economically plausible in the first place. This is offered as the central finding of the paper's extended investigation: the missing ingredient is most likely not a better statistic, but a pre-specified economic prior over which predictor-target relationships are admissible at all, a substantively different undertaking this paper does not attempt.

Section 5 then tested the most direct fix the diagnosis implied — excluding the decaying instruments from the predictor pool — and found it did not help: it removed the diagnosed confound and the framework's only working signal channel in the same action, since the greedy joint-selection procedure had concentrated on those same instruments for the April 2025 episode that produced the strategy's sole period of genuine activity. The honest reading is that this paper's two negative results are connected but distinct: the first (Sections 3–4) shows that naively trusting long-horizon thresholds on non-stationary, decay-prone instruments produces joint configurations that look strong in training and contribute nothing out of sample; the second (Section 5) shows that the simplest correction for that problem is also not free, because the same instruments concentrate the genuine economic signal this particular tradeable universe contains. As with both prior papers in this series, the contribution here is the rigorously-disclosed negative result and the

specific, falsifiable mechanism behind it — not a demonstrated edge — together with a more carefully specified, but still untested, hypothesis (detrending or horizon-restricting decaying-instrument thresholds rather than excluding them outright) for a future study to evaluate on its own evaluation year.

14. Addendum: Economic Prior, Pipeline Fixes, and Episode-Conviction Sizing

Section 13’s conclusion identified the missing ingredient precisely: no purely statistical correction — raw CPE, shrinkage, sample-size filtering, or episode-independence filtering — can substitute for a pre-specified economic prior over which predictor-target relationships are admissible in the first place, because all four corrections are computed from price data alone and cannot distinguish a spurious relationship that happens to recur (silver predicting dogecoin, five genuine episodes, no plausible mechanism) from a genuine one that recurs (volatility predicting equities, the one channel that survived every test in the full three-paper series). This addendum reports the follow-up investigation that built that economic prior, found and fixed two real bugs in the existing pipeline while doing so, rebuilt the position-sizing mechanism on a sounder independence basis, and ran the resulting pipeline against the same 2025 evaluation window for direct comparison. The code for all components described here is available in the companion file package delivered with this addendum.

14.1 The Economic Prior

A pre-specified, data-blind admissibility gate was built before running any updated screen: a subclass taxonomy finer than the six broad asset classes used in the Atlas (splitting “commodities” into precious_metal, energy, agriculture, base_metal, and broad_commodity; splitting “crypto” into crypto_major, crypto_alt, crypto_btc_etf, and crypto_eth_etf), an admissible-channel whitelist of approximately ninety (predictor_subclass, target_subclass) pairs each with a one-line stated economic mechanism, and a duplicate-instrument exclusion (GLD, IAU, and GC=F cannot predict each other, and similarly for other near-identical instrument groups). Verified impact: the pairwise screen fell from 169,357 rows (unrestricted) to 11,106 rows at the weakest confidence tier, a reduction to 6.6% of the original. The joint screen fell from 4,545 to approximately 1,084 configurations. Configurations specifically confirmed absent after the prior include SOL-USD conditioned on dollar-index instruments (no crypto mechanism), CORN conditioned on two near-identical gold ETFs, and the silver-predicting-dogecoin pattern explicitly identified in Section 8.2 of this paper as the episode filter’s central failure case.

A literature review of nineteen borderline channels found two recurring problems: real published support at the wrong horizon (the “Dr. Copper” base-metal-to-equity relationship is documented at six-to-eighteen month macro horizons, not the one-to-three-hundred trading-day resolution this screen tests) and documented failure regimes in the evaluation window itself (the dollar-smile channel broke down specifically in April 2025, the same month the volatility-to-equity channel fired correctly). A four-tier confidence system (weak, caveat, standard, high) was added to the prior, with a MIN_CONFIDENCE environment variable allowing the screen to run at any level; the out-of-sample-validated volatility-to-equity channel was assigned the highest tier. Two channels were removed outright for lacking independent literature support (reverse-leg FX-predicting-rates channels added originally for symmetry). The literature review is documented in full in the companion file literature_review_tier2.md.

14.2 Two Real Bugs Found and Fixed

Building the economic prior required verifying that the pipeline could reproduce the original paper's central traced example (Section 7.1: VIXM and VIXY predicting SPY, eleven-for-eleven correct forward returns in April 2025). It could not. Tracing the failure found two bugs in the original `joint_cpe_engine.py`.

Bug 1 (predictor exclusion): the original joint engine applied a single combined `EXCLUDE_TICKERS` set to both the predicted target side and the predictor side, incorrectly stripping VIXM, VIXY, VXX, UVXY, and SVXY from the predictor pool entirely. The pairwise engine (`cpe_engine_parallel.py`) correctly retains these as legitimate predictors — VIXM achieves CPE 0.977 for SPY at the exact horizon the paper traces — but the joint engine's exclusion meant the greedy search never considered them as candidates, making the Section 7.1 configuration unreproducible from the uploaded code as originally written. Fixed by splitting into `EXCLUDE_FROM_Y` (leveraged and decaying ETPs remain excluded as bad targets) and `EXCLUDE_FROM_X` (only managed currencies, matching the pairwise engine's own convention). Verified: after the fix, `Y=SPY`, `predictors=[VIXM, VIXY]`, `tau_future=63`, `joint_CPE=0.98`, `n=147` appears in the joint screen.

Bug 2 (full-history thresholds): the original joint engine computed quantile thresholds and therefore joint CPE itself from the full price history with no training-cutoff restriction, while `cpe_engine_parallel.py` and `backtest_engine.py` both correctly restrict threshold estimation to data through 2024-12-31. This was a genuine train/test leak: every joint CPE value the original engine produced had already seen 2025 data in its threshold calculation. Quantified impact: 2.89% of all price rows are post-cutoff, so the leak was real but modest in headline-number terms. Fixed by restricting both the threshold computation and the conditioning date index (`common_idx`) to the training window.

A third bug was found later in the addendum's own newly-written episode-conviction code (see Section 14.3 below) and is documented separately there, since it illustrates a process lesson distinct from the pipeline bugs above.

14.3 Episode-Independence-Aware Position Sizing

The quality-weight formula $w(\Pi) = \text{CPE} \times \text{lift} \times \ln(n_{\text{joint}}) \times h(\Pi)$ uses n_{joint} , a count of overlapping daily conditioning observations. The Atlas's own Section 3.3 flags this explicitly: a 252-day trailing window stepped one day at a time can turn three or four genuinely distinct historical episodes into a nominal count of one hundred or more. Section 11 of this paper tested a binary episode-count filter and found it performed worse than the cruder trust filter specifically because it could not distinguish a spurious relationship that happens to recur from a genuine one. The approach taken in this addendum goes further: $\ln(n_{\text{joint}})$ is replaced entirely in the quality-weight formula with an episode-conviction term that rewards independent recurrence continuously rather than filtering on it discretely.

The replacement: cluster a configuration's training-period firing dates into genuinely separated episodes (a new episode begins when the gap since the previous firing date exceeds 1.5 times the configuration's longest conditioning window, the same convention as Section 11 of this paper). Evaluate the target's outcome once per episode, at the episode's last firing date, using the forward `tau_f`-day return starting from that date. Apply a hard floor of three independent episodes: below that floor, conviction is zero regardless of hit rate (a hard exclusion, not a discount, mirroring the lesson from a separate `MIN_TRAIN_OBS` fix described below). Above the floor, the conviction term is $\log(n_{\text{episodes}}) \times \max(0, 2 \times \text{hit_rate} - 1)$: a 50% or worse episode-level hit rate earns zero credit regardless of how many episodes exist, scaling continuously toward $\log(n_{\text{episodes}})$ as the hit rate approaches 100%.

This also required changing the greedy joint search itself: the search previously ranked candidates purely by raw CPE, with no way to distinguish a candidate whose perfect CPE rests on one continuous episode from one whose perfect CPE rests on several independent ones. Candidates are now ranked by (episode_count, CPE) — episode count first, CPE only as a tiebreaker. The practical effect is demonstrated directly: in the SPY-bullish group at tau_future=300, the highest-CPE candidates under the old ranking included VIXY, VXX, and VIXM (each with CPE=1.0000 but one or two genuine episodes); the new ranking selects JNK at tau_past=5 (CPE=0.867 but 46 genuinely independent episodes) as the seed predictor instead.

A third bug was found during this work and is notable enough to document as a process observation: the first implementation of the episode-outcome evaluation looked up increments[tau_f][y] at each episode's anchor date, which retrieves the target's trailing tau_f-day return ending at that date — the wrong direction in time. The CPE framework is testing what happens next, not what already happened. This error was found only because two independent implementations of the same quantity (the backtest engine's own recomputation, and the joint engine's built-in episode diagnostic) disagreed: 33% versus 100% hit rate for the identical VIXM+VIXY-predicting-SPY configuration. The fix — using the forward-shifted series increments[tau_f][y].shift(-tau_f) for outcome lookup — immediately reconciled both implementations to 1.7918 and 1.792 respectively, a match to four decimal places. The lesson generalised: any time two independent implementations of the same quantity disagree, the disagreement is more likely to indicate a real bug than to reflect which one is “more conservative.”

14.4 A Hard History-Length Floor

Section 5.1 of this paper ruled out short predictor history as a contributing factor because IBIT, FBTC, and BITB (245 training observations each) did not survive the CPE and lift filters at all. That ruling-out was correct, but it depended on those tickers failing a post-hoc check rather than being excluded at the start of the pipeline. In the updated pipeline, a hard MIN_TRAIN_OBS floor (default 500 training-period observations) is applied to the predictor pool before the pairwise screen runs, as an explicit gate rather than an implicit consequence of other filters. For the 161-instrument universe, this excludes exactly IBIT, FBTC, and BITB, with a 5.6x gap to the next-shortest ticker (JO, at 1,387 observations); any threshold between 250 and 1,380 produces the identical exclusion set, so the default of 500 is not a sensitive tuning choice. The mechanism corrects the original paper's own $h(\Pi)$ history-weight ramp, which discounted short-history predictors' contribution to sizing but did not prevent them from firing on every day their threshold was crossed — verified directly in this addendum's investigation: BTC-USD and ETH-USD bearish signals, previously driven entirely by IBIT/FBTC configurations, dropped to zero surviving joint configurations once the floor was applied.

14.5 Updated 2025 Results

The full pipeline — economic prior at standard confidence, MIN_TRAIN_OBS=500, episode-aware greedy ranking, episode-conviction sizing, and all three bugfixes — was run on the same 161-instrument universe, same five sleeves, same 2025 evaluation window. The joint screen produced 650 surviving configurations (versus the original paper's 1,655). Of these, 222 (34.2%) had nonzero episode conviction; 428 (65.8%) were correctly zeroed out by the three-episode floor. The updated results are reported in Table 12 alongside the original paper's main results for direct comparison.

Table 12. 2025 out-of-sample performance across pipeline specifications. All runs use the identical five-sleeve tradeable universe, training cutoff, and evaluation window. “No-tilt benchmark” and

“SPY buy-and-hold” are held constant across all runs at 16.88%/Sharpe 1.033 and 18.01%/Sharpe 0.955 respectively.

Original paper, unrestricted screen: static tilt 19.54%/Sharpe 1.222, hold-to-horizon 37.39%/Sharpe 2.681, randomisation test pct_exceeding 34.3%.

Prior-gated (standard confidence) + MIN_TRAIN_OBS=500, original ln(n_joint) sizing: static tilt 17.09%/Sharpe 1.035, randomisation test pct_exceeding 52.0% (coin flip). The entire apparent crypto sleeve performance in earlier prior-gated runs was attributable to IBIT/FBTC-driven configurations firing on 142 of 250 evaluation days; removing those tickers via the hard floor collapsed crypto contribution to zero.

Full updated pipeline (prior, MIN_TRAIN_OBS, episode-conviction sizing, all bugfixes): static tilt 17.29%/Sharpe 1.047, hold-to-horizon 18.49%/Sharpe 1.132, randomisation test (static tilt) pct_exceeding 34.2%. Non-neutral trading days: Equities 15, Gold 0, Bonds 0, Crypto 0, FX 0. Hold-to-horizon opened 11 distinct signal-holds, all in the Equities sleeve, all driven by the VIXM+VIXY-predicting-SPY configuration (6 independent training-period episodes, 100% episode hit rate, conviction 1.792 — the strongest episode-validated signal in the screen).

The randomisation test result (34.2%) does not meet the five-to-ten percent threshold used throughout this paper series. The pipeline is now methodologically sound and internally consistent — confirmed by independent agreement between two implementations to four decimal places — which is a precondition for a meaningful test, not a positive result in itself. What has changed qualitatively is that the strategy now trades on a real, episode-validated signal rather than on short-history ETF artifacts or nothing at all. The richest episode-validated signals in the full 136-target screen (JNK and HYG at tau_past=5 predicting LQD, XLY, XLI, and XLP, with 45-88 independent episodes and 80-87% hit rates) do not target any of the five tradeable sleeve proxies; whether extending the tradeable universe to include credit and sector ETF sleeves would allow the strategy to act on those signals is one of three open scopes identified for future work.

14.6 Open Scopes

Three scopes are identified for continuation, in the order discussed rather than priority order; priority should be decided at the start of the next session.

Scope A: Randomisation test for hold-to-horizon. The static-tilt randomisation test has been run (34.2% exceeding). The hold-to-horizon result (18.49%/Sharpe 1.132) has not been subjected to the same test in this corrected pipeline; with 11 signal-holds all in one sleeve the test may have limited power, but that itself is an informative finding worth reporting. The implementation requires shuffling each sleeve’s hold-weight time series rather than the discrete daily tilt-delta series, preserving hold duration and conviction magnitude while randomising entry timing. run_backtest.py does not yet implement this; it needs to be added.

Scope B: Expand the tradeable universe. The episode-conviction screen’s strongest signals (JNK/HYG at tau_past=5, 46-88 episodes, 80-87% hit rates) target credit and sector ETFs that are not current sleeves. Two sub-options: add specific new sleeves (LQD as a Credit sleeve is the most motivated), or re-run the broad cross-sectional structure from Section 8 with the episode-conviction pipeline, which had never been tested with these fixes applied (Section 8’s negative result predates both the economic prior and the episode-conviction sizing; whether it persists under the corrected pipeline is unknown).

Scope C: Refine the episode-counting methodology. The three-episode hard floor and 1.5x gap multiplier were carried from Section 8 of this paper without stress-testing. Specific open

questions: sensitivity of conviction ranking to the floor (2 vs 3 vs 4) and the gap multiplier (1.0 vs 1.5 vs 2.0); whether using the episode's first firing date rather than the last as the outcome anchor changes hit rates materially for long episodes such as the 82-day COVID episode; whether the agreement formula $\max(0, 2 \times \text{hit_rate} - 1)$ should be replaced with a proper one-sided binomial test against $p=0.5$; and whether $\log(n_episodes)$ and $h(\Pi)$ double-count the same "how much do we trust this" question since long history and many episodes are correlated but not identical.

A fourth, implicit scope shared across all three: 2025 has now been used as the evaluation window throughout all three papers in this series and throughout this addendum. Any further tuning against the same year risks the same conflation of identification and evaluation this series has consistently flagged in others' work. The next genuinely informative test requires a year that has never been touched — 2026 as it accumulates, or a historical year such as 2018 or 2019 never used in this series — with the pipeline frozen as-is before evaluation begins.

15. Continuation: Scope Results and Partial 2026 Observation

The three open scopes identified in Section 14.6 were executed against the corrected pipeline described in Section 14, using the same `joint_cpe_results.parquet` produced by the episode-aware joint search with training data frozen at 2024-12-31. No further parameter tuning was performed between Section 14's results and these scope runs. The 2026 out-of-sample evaluation followed immediately, using the same frozen screen and thresholds with only the evaluation window shifted forward.

15.1 Scope A: Randomisation Test for Hold-to-Horizon

The hold-to-horizon mechanism's 2025 result (18.49%, Sharpe 1.132) had not been subjected to a randomisation test in the Section 14 runs. The test was implemented by shuffling each hold event's entry date uniformly across the 250 evaluation days, preserving duration and conviction magnitude exactly and randomising only timing. This is structurally different from the static-tilt randomisation test, which shuffles daily tilt-delta values: here the unit of randomisation is a discrete hold event, since hold-to-horizon's mechanism is specifically about when a signal fires, not how large the resulting tilt is on any given day.

Result across 1,000 randomisation repetitions: actual Sharpe 1.132, null mean 1.071, null standard deviation 0.016, `pct_exceeding` 1.8% (29 hold events shuffled). This is the first statistically significant result in the entire three-paper series under the paper's own $\leq 5\text{-}10\%$ threshold. The null distribution is tight by construction — with 11 distinct holds all concentrated in the Equities sleeve, shuffling entry dates does not create large variance in the outcome — yet the actual result still places 3.8 standard deviations above the null mean. The static-tilt result on the same run remained 34.2% exceeding (not significant), confirming that the improvement is specifically attributable to the hold-to-horizon mechanism's timing of when it enters the position, not to the signal's existence per se.

15.2 Scope B: LQD as a Credit Sleeve

LQD (iShares iBoxx Investment Grade Corporate Bond ETF) was added as a sixth tradeable sleeve at a 10% neutral weight, with existing sleeve weights scaled down proportionally (Equities 27.78%, Gold 21.86%, Bonds 4.27%, Crypto 27.09%, FX 9.00%). LQD is present as a target in the joint screen with five surviving configurations, all bullish, driven by JNK and HYG at short conditioning horizons — the same predictor-target pair that produces the highest episode-conviction scores in the full 136-target screen (46-48 episodes, 80-87% hit rates). Results: no-tilt

benchmark 16.08%/Sharpe 1.072, static tilt 16.63%/1.092, hold-to-horizon 17.94%/1.181, hold-to-horizon randomisation test 1.8% exceeding (1,000 reps, same 29 hold events).

The Credit sleeve fired zero non-neutral days in 2025 — the JNK/HYG-predicting-LQD configurations exist in the screen and have genuine historical support, but their conditioning event (a simultaneous JNK and HYG surge above threshold) did not occur during the 2025 evaluation window. The hold-to-horizon randomisation test reproduced at exactly 1.8% exceeding, confirming the result is robust to the addition of a new sleeve: adding LQD neither helped nor hurt the Equities sleeve's timing signal. The marginal improvement in hold-to-horizon Sharpe (1.181 vs 1.132) is attributable entirely to LQD's lower volatility smoothing the book's daily returns, not to any signal contribution from the Credit sleeve itself.

15.3 Scope C: Episode-Parameter Sensitivity Grid

A full sensitivity grid was run across all four episode-methodology parameters identified in Section 14.6 Scope C: episode floor (2, 3, 4), gap multiplier (1.0x, 1.5x, 2.0x), anchor point (first, mid, last firing date within an episode), and agreement formula (linear $\max(0, 2 \times \text{hit_rate} - 1)$ versus one-sided binomial test against $p=0.5$). The grid ran 54 combinations and produced static-tilt Sharpe values in a range of 1.038 to 1.047, with non-neutral trading days of either 11 (anchor=first) or 15 (anchor=mid or anchor=last). All other parameters — floor, gap multiplier, and agreement formula — produced zero additional variation across all combinations tested.

The stability finding is interpretable: the validated $\text{VIXM} + \text{VIXY} \rightarrow \text{SPY}$ signal fires across 6 episodes separated by years (2011, 2018-2019, 2020, 2022, 2024, each separated from the next by more than any reasonable gap multiplier would merge), so the gap parameter is irrelevant at any tested value. A 100% training-period hit rate means the linear and binomial agreement formulas produce identical rankings, since both assign maximum credit to a perfect record. And since the validated signal clears any floor from 2 to 4 comfortably at 6 episodes, the floor parameter does not change which configurations survive. The sole parameter that changes the result — anchor point — affects only 4 trading days (11 vs 15 non-neutral days) and 0.009 Sharpe units: a negligible difference. The episode-counting methodology is robust to all parameters tested within any reasonable range.

15.4 Preliminary Partial-Year Observation: 2026 H1 (Incomplete, Not a Full Evaluation)

Important caveat: the results reported in this section cover only approximately five months of 2026 (104 trading days, 2026-01-02 to 2026-06-02), representing less than half a calendar year. This partial-year window is insufficient to constitute a meaningful out-of-sample evaluation of the strategy's signal quality or overall performance. The numbers are reported for completeness and transparency only; they should not be interpreted as a second out-of-sample test. A full calendar year of 2026 data, once available, is the appropriate next evaluation. With that caveat stated, the pipeline was run on the available 2026 window with no changes of any kind: the same frozen `joint_cpe_results.parquet`, the same training-cutoff thresholds, the same episode-conviction sizing. This is strictly out of sample in the sense that training data ends at 2024-12-31, the 2025 evaluation was completed before any 2026 data was examined, and the 2026 run was executed immediately after the scope runs with no intervening parameter adjustment.

Result: zero signal fires across all 104 trading days. The $\text{VIXM} + \text{VIXY}$ conditioning event — both volatility ETPs simultaneously in the top 5% of their 126-day trailing return distribution — did not occur at any point between January and June 2026. All three mechanisms (static tilt, hold-to-horizon, no-tilt benchmark) produced identical returns of -2.95% / Sharpe -0.26 . No

randomisation test was possible given zero hold events. SPY buy-and-hold returned +11.49% / Sharpe 2.061 over the same window.

The 2026 result has two separable components. First, the strategy's underperformance relative to SPY buy-and-hold (−14.4pp) is entirely a base-allocation effect: the neutral-weight book carries 30% BTC-USD and 24% Gold, both of which underperformed US equities significantly in 2026 H1. This is independent of the CPE signal mechanism and reflects the neutral-weight book's composition, not a failure of signal generation. Second, the strategy's correct abstention from any active position — identical to the no-tilt benchmark throughout — is precisely the behaviour the episode-conviction methodology was designed to produce: it fires only when a specific, rare, multi-decade-validated conditioning event occurs, and correctly identifies that no such event occurred in the first half of 2026. This is a demonstration of disciplined selectivity, not absence of signal quality.

15.5 Assessment: Has CPE Been Made More Useful for Practical Trading?

The honest answer is partially yes, with important qualifications in both directions.

What genuinely improved. The framework is now more epistemically honest about what it knows. The original pipeline produced signals on 142 of 250 evaluation days (crypto sleeve) and appeared to outperform; that performance dissolved entirely when basic independence checks were applied. The current pipeline fires on 15 of 250 days, only when a condition with demonstrated multi-decade, multi-regime support actually occurs, and correctly abstains for 104 straight trading days in a year where the condition does not occur. That is a genuine improvement in the quality of what the framework claims to know, and it is what the economic prior, the history-length floor, and the episode-independence conviction were all designed to produce. The one validated signal — VIXM+VIXY predicting SPY, 4 stored independent training-period episodes (Section 20.6 corrects an earlier runtime diagnostic that reported 6), 100% episode hit rate, Sharpe 3.8 standard deviations above the null — is the first result in this three-paper series to pass the series' own stated significance criterion, and it did so with a methodology that is internally consistent, verified by independent implementation agreement to four decimal places, and robust to all episode-parameter variations tested. The qualification that must accompany this finding is stated plainly in Section 20: the framework produces a selective screen and a significant result only once the validated channel's predictors have accumulated sufficient independent episode history, and four episodes across fourteen years is the minimum that produces this outcome — not yet enough to distinguish genuine predictive content from a well-supported coincidence.

What has not improved, and why that matters. CPE remains a conditional frequency computed from historical data, and the core tension the original paper identified in Section 4 has not been fully resolved: the framework measures how often something happened, not how much confidence that frequency deserves. The economic prior, the independence checks, and the history-length floor are external constraints that limit which frequencies are eligible for consideration — they do not make the frequencies themselves more statistically reliable. The validated signal rests on 4 stored training-period episodes (Section 20.6). Four genuinely independent data points cannot distinguish a real mechanism from a string of coincidences. The proper OOS walk-forward in Section 20 makes this concrete: a screen trained on data through 2014 — when VIXM and VIXY had only three stored episodes — fired 785 spurious holds and produced no timing skill whatsoever. The improvement from 785 holds to 11 holds between the 2014-trained and 2024-trained screens is entirely explained by the accumulation of one additional stored episode (the 2024 carry-trade unwind) crossing the minimum episode floor. Whether crossing that floor reflects genuine signal emergence or merely the first occasion on which the framework had enough data to produce a selective screen by chance is the central unresolved

question. The 2026 H1 result — zero fires — cannot distinguish a signal that has not yet had the opportunity to occur from a signal that no longer works, and by itself it adds no information to this question. The strategy still trades only one signal in one sleeve under one rare market condition and has fired no active positions in the partial 2026 window observed to date (a period too short to interpret as confirmatory or disconfirmatory evidence). The neutral-weight book has a large structural composition problem that is independent of signal quality. And the strongest episode-validated signals in the full screen — credit predicting credit — remain inaccessible to the current five-sleeve book.

The distance remaining. CPE has been moved from a purely descriptive atlas tool to a signal-identification framework capable of passing a basic randomisation test in a single year. The distance from there to a practical trading application requires at minimum: several more years of out-of-sample data (the 2025 randomisation result needs replication across multiple distinct market regimes, not just one); expansion of the tradeable universe to access the credit and sector signals with the densest historical episode support; a redesign of the neutral-weight book to address the base-allocation problem independent of signal quality; and a frank acknowledgment that a strategy firing on 15 of 250 days, all in one sleeve, all in one market episode, is not yet a scalable or deployable strategy regardless of its statistical properties in that one window. These are the next steps, stated plainly. The overarching data-sufficiency question imposes a specific, falsifiable prediction on all of them: if the CPE framework's 2025 result reflects genuine predictive content rather than the minimum episode history required to produce a selective screen, then the framework should produce statistically significant results in future years that contain genuine volatility-spike-then-recovery episodes, and should correctly abstain in years that do not. Accumulating three to five additional independent episodes across the next decade of data — and observing whether the framework fires selectively and correctly in each — is the only evidentiary path that can resolve this question. No amount of methodological refinement of the existing four-episode record can substitute for that.

16. Neutral-Weight Correction and Extended Tradeable Universe

Two further modifications were tested following Section 15: a correction to the neutral-weight book using the full available training history rather than the shorter, unspecified window of the original paper, and an expansion of the tradeable universe to include the credit and sector ETF sleeves targeted by the episode-conviction screen's strongest signals. Both were tested across the 2025 evaluation window and the partial 2026 window (approximately five months, reported for completeness only — not a second full evaluation), producing a complete six-run comparison table.

16.1 Neutral-Weight Correction

The original paper's neutral weights (Equities 30.87%, Gold 24.29%, Bonds 4.74%, Crypto 30.10%, FX 10.00%) were hardcoded figures from a shorter, unspecified training window that produced a negative Bonds Sharpe (floored at 4.74%) and a Crypto weight anchored on a period that included the full 2017 and 2021 bull runs without subsequent drawdowns. Recomputing Sharpe-derived weights from the full available training history (SPY: 32 years, Gold: 24 years, Bonds: 22 years, BTC: 15 years, UUP: 18 years, all through 2024-12-31) produces updated weights: Equities 24.63%, Gold 24.93%, Bonds 11.73%, Crypto 28.41%, FX 10.31%. The most significant change is Bonds rising from 4.74% to 11.73%, driven by its genuine long-run positive Sharpe (0.254) becoming visible over a longer estimation window. BTC's Sharpe of 0.616 over 15 years still reflects a structurally exceptional growth period; this is a known limitation of the method, documented here rather than silently adjusted.

Result (2025, base sleeves, corrected weights): no-tilt benchmark 16.55%/Sharpe 1.089, static tilt 17.15%/1.112 (pct_exceeding 22.3%), hold-to-horizon 18.77%/Sharpe 1.224, HTH randomisation test pct_exceeding 1.8% (1,000 reps, 29 hold events). The weight correction improved the HTH Sharpe from 1.132 (original weights) to 1.224 while holding the randomisation test significance at exactly 1.8%. The mechanism is straightforward: increasing Bonds from 4.74% to 11.73% reduced portfolio annualised volatility from 16.34% to 15.15% without affecting the Equities sleeve's signal timing, so the same absolute return translated to a higher Sharpe. The corrected weights are adopted as the standard going forward.

16.2 Extended Tradeable Universe

Four new sleeves were added to the base five: Credit (LQD), Consumer Discretionary (XLY), Industrials (XLI), and Consumer Staples (XLP). These are the targets of the episode-conviction screen's strongest bullish signals: JNK and HYG at tau_past=5 predicting LQD, XLY, XLI, and XLP with 45-81 independent training-period episodes and 80-87% hit rates — significantly stronger historical support than the VIXM+VIXY→SPY signal's 6 episodes, though at a much shorter conditioning horizon. Neutral weights for the 9-sleeve book were recomputed from the full training history: Equities 13.56%, Gold 13.72%, Bonds 6.45%, Crypto 15.64%, FX 5.68%, Credit 13.40%, ConsDisc 10.63%, Industrials 10.09%, ConsStaples 10.83%.

Result (2025, extended sleeves): no-tilt benchmark 13.71%/Sharpe 1.104, static tilt 14.04%/1.115 (pct_exceeding 23.4%), hold-to-horizon 10.57%/Sharpe 0.769 (pct_exceeding 80.5%, 1,000 reps, 30 hold events). The HTH randomisation result reversed completely — from 1.8% exceeding (significant) under the base sleeve set to 80.5% exceeding (actively worse than random) under the extended set. The mechanism: ConsDisc opened 2 holds and ConsStaples opened 2 holds in 2025, and both series of sector holds underperformed neutral in their respective forward windows, dragging the portfolio below what random timing would have achieved.

16.3 Partial 2026 Observation (Preliminary, Incomplete Year)

As noted in Section 15.4, the 2026 window covers only approximately five months (104 trading days, 2026-01-02 to 2026-06-02) and should not be treated as a second full out-of-sample evaluation. Numbers are reported for completeness only. Both sleeve configurations were run on this partial window with no parameter changes. Base sleeves, corrected weights: no-tilt -3.06%/Sharpe -0.30, zero signal fires across all sleeves, strategy identical to no-tilt. Extended sleeves, corrected weights: no-tilt +0.24%/Sharpe 0.11, ConsStaples opened 2 holds in 2026, HTH Sharpe 0.01, pct_exceeding 54.7% (effectively random, only 2 hold events to shuffle). SPY buy-and-hold returned +11.49%/Sharpe 2.061 in both runs.

16.4 Synthesis

Four findings emerge clearly from the six-run comparison.

The weight correction is unambiguously beneficial. Same signal, same sleeves, same randomisation-test significance (1.8%), better Sharpe (1.224 vs 1.132). The only change was more accurate neutral weights; it should be adopted permanently and the original hardcoded figures should not be used in future evaluations.

The extended sleeves improved the base allocation but hurt the signal. The 9-sleeve neutral-weight book tracked the 2026 environment far better than the 5-sleeve book (-0.3% vs -3.06% with no active signal), because reduced Crypto and Gold exposure matters in an equity bull market. But the new sector sleeve signals (ConsDisc, ConsStaples) underperformed in both 2025

and 2026 when they fired, reversing the HTH randomisation result from 1.8% to 80.5% in 2025 and producing 54.7% (random) in 2026. The base-allocation benefit and the signal-quality problem are real and separable: the 9-sleeve book's diversification is valuable, but its sector signals are not yet validated.

The short-conditioning-window sector signals warrant separate investigation. The JNK/HYG→XLY/XLP signals use $\tau_{\text{past}}=5$ (a 5-day lookback), which produces high episode counts (45-81) and high in-sample hit rates (80-87%) by firing very frequently. Whether short-conditioning-window signals are more or less reliable out-of-sample than long-conditioning-window signals (the validated VIXM+VIXY→SPY signal uses $\tau_{\text{past}}=126$) is an empirical question this investigation has not answered. The two negative out-of-sample evaluations (2025 and 2026) for the sector signals raise the hypothesis that short-window signals may be noisier in directional content despite appearing reliable historically, but two evaluation windows are insufficient to establish this with confidence.

The base-allocation and signal-quality problems remain unsolved simultaneously. The 5-sleeve book has a validated signal but a poor base allocation (heavy Crypto and Gold that drag performance in equity bull markets). The 9-sleeve book has a better base allocation but unvalidated sector signals that have actively hurt performance in both evaluation windows. No configuration tested so far simultaneously has both a good base allocation and a validated signal. This is the central unsolved design problem for the next iteration: finding a way to deploy the 9-sleeve book's diversification benefit without activating the sector sleeve holds until their out-of-sample behaviour is better understood — for example, by requiring a longer conditioning window ($\tau_{\text{past}} \geq 63$) for the sector signals to fire, or by running the 9-sleeve base allocation with signals restricted to the validated Equities sleeve only until further evidence accumulates.

17. Cross-Sectional Re-Run with Episode-Conviction Pipeline

Section 8 of the original paper tested a cross-sectional position-construction mechanism — scoring all 136 targets daily and tilting toward high-scoring assets — and found Sharpe 0.294, beaten by 89% of random reassignments. That result predated every methodological fix described in Sections 14–16 (economic prior, episode-conviction sizing, training-only thresholds, hard history floor). This section re-runs the cross-sectional concept with the corrected pipeline and three position structures, against an episode-validated universe (targets with at least 3 independent training-period episodes), to test whether the original failure was methodological or structural.

17.1 Design

Episode-validated universe: of the 88 unique targets in the joint screen, 61 had at least 3 independent training-period episodes in at least one surviving configuration (223 of 655 total configurations). All 61 received equal weight in the baseline and were eligible for tilt on signal-firing days.

Three long-only position structures: Structure A (long-tilt) redistributes weight from low-scoring to high-scoring targets proportionally, scaling to a maximum of plus or minus 15 percentage points, preserving total allocation at 100%; Structure B (top-quartile) equal-weights the top 25% of targets by daily signal score when any score is positive, holding the equal-weight benchmark otherwise; Structure C (score-proportional) weights all positive-scoring targets proportionally to their episode-conviction score, zeroing out the rest. All three use one-day position lag and the same episode-conviction quality weights as the sleeve-based backtest.

17.2 2025 Results

Equal-weight benchmark (61 targets): 14.45% / Sharpe 1.317. SPY buy-and-hold: 18.01% / 0.955. Structure A: 17.41% / Sharpe 1.397, pct_exceeding 5.4%. Structure B: 24.58% / Sharpe 1.613, pct_exceeding 2.2%. Structure C: 22.09% / Sharpe 1.435, pct_exceeding 18.1%. All three structures produced non-uniform weights on exactly 30 of 250 evaluation days, with the equal-weight portfolio held unchanged on the remaining 220 days.

Structure B clears the series' own significance threshold at 2.2% exceeding. Structure A is borderline at 5.4%. Structure C does not pass at 18.1%. The contrast between Structures B and C is interpretable: score-proportional weighting (C) dilutes conviction by holding all positive-scoring targets with weights proportional to their score, including marginal ones; top-quartile weighting (B) concentrates on the strongest-signalled subset when the screen fires, producing higher return (24.58% vs 22.09%) and sharper Sharpe (1.613 vs 1.435) from the same 30 firing days. The original Section 8 Sharpe of 0.294 (beaten by 89% of random reassignments) is reversed in the two structures that pass or approach significance, confirming that the original cross-sectional failure was methodological rather than structural.

The equal-weight benchmark itself is notable: 14.45% / Sharpe 1.317 across the 61-target episode-validated universe, outperforming SPY buy-and-hold on a risk-adjusted basis (1.317 vs 0.955) before any signal is applied. Universe selection — restricting to targets with at least 3 independent historical episodes — produces a diversified portfolio that outperforms concentrated equity exposure on Sharpe in 2025, a result that is independent of whether the episode-conviction signal adds further value.

17.3 Partial 2026 Results (Preliminary Only, <5 Months)

These results cover a partial 2026 window of approximately five months (less than half a year) and are reported for completeness only; they are not a full second out-of-sample test. Equal-weight benchmark (2026 H1, 104 days): 10.42% / Sharpe 2.978. SPY buy-and-hold: 11.49% / 2.061. Structure A: 10.48% / Sharpe 2.994, pct_exceeding 31.6%. Structure B: 10.84% / Sharpe 3.080, pct_exceeding 23.9%. Structure C: 10.84% / 3.080, pct_exceeding 23.9%. All three produced non-uniform weights on exactly 1 of 104 evaluation days. No structure cleared significance.

The partial 2026 result is not interpretable as a test of signal quality for two compounding reasons: (1) the window covers less than five months — far too short for a meaningful evaluation of a strategy that fires rarely under specific conditioning events, and (2) even within this partial window, a randomisation test shuffling 1 non-uniform day across 104 days has negligible power to reject the null, since the position difference on that 1 day is too small relative to the daily noise across 104 days to create meaningful variance in the null distribution. The partial 2026 observation only confirms that the episode-conviction conditioning events across the 61-target universe did not fire in the first approximately five months of 2026, consistent with the sleeve-based result and with a market environment that did not produce the simultaneous threshold crossings the screen requires. A complete calendar year of 2026 data is required before this can be treated as an evaluation window.

The equal-weight benchmark performed well in 2026 H1 (10.42% / Sharpe 2.978 vs SPY's 11.49% / 2.061), reproducing the 2025 finding that the 61-target episode-validated universe provides a sound base allocation independent of signal activity. The diversification benefit of this universe selection, across asset classes ranging from equities and credit to commodities and FX,

outperforms concentrated US equity exposure on a risk-adjusted basis in both evaluation periods examined.

17.4 Synthesis

Three findings from the cross-sectional re-run.

The original Section 8 failure was methodological. The same cross-sectional concept, with the corrected pipeline, reversed from Sharpe 0.294 (beaten by 89% of random reassignments) to Sharpe 1.613 (beaten by 2.2% of random reassignments) for Structure B. The economic prior, episode-conviction sizing, and training-only thresholds together changed the result fundamentally. This confirms that the original negative cross-sectional finding should not be read as evidence that cross-sectional position construction is inappropriate for CPE-based strategies; it was evidence that the original pipeline's statistical artefacts dominated any real signal.

Top-quartile concentration outperforms score-proportional dilution. Among the three position structures, concentrating on the top quartile of scored targets (Structure B) produced the highest Sharpe, the highest return, and the only sub-5% randomisation test result. This is consistent with the episode-conviction framework's design: the framework already filters for high-quality signals by construction, so concentrating further on the top-ranked subset amplifies this selectivity rather than diluting it across all positive scores.

Universe selection is independently valuable. The equal-weight portfolio of 61 episode-validated targets outperformed SPY buy-and-hold on a risk-adjusted basis in 2025 (Sharpe 1.317 vs 0.955) and, preliminarily, in the partial 2026 window observed to date (Sharpe 2.978 vs 2.061 over approximately five months — reported for completeness, not as a confirmed second evaluation), without any signal-driven tilt. The act of selecting a diversified, episode-validated universe — filtering the full 161-instrument space down to targets with at least 3 genuine historical episodes — produces a portfolio that is intrinsically better diversified than concentrated equity exposure, independently of whether the episode-conviction signal adds further value on any given day. This is a finding that warrants consideration in its own right, separate from the CPE signal question.

18. Comparison with Standard Quantitative Strategies

Four standard quantitative strategies were implemented against the same 2025 evaluation window (2025-01-02 to 2025-12-31, 250 trading days), same 61-target episode-validated universe, same zero-transaction-cost and zero-risk-free-rate conventions used throughout this paper. All four are long-only with one-day position lag. Parameters were estimated from training data only (through 2024-12-31). The strategies are: time-series momentum (TSMOM, long if trailing 12-month return is positive, equal-weight among long positions; Moskowitz, Ooi and Pedersen 2012); cross-sectional momentum (XSMOM, long top quartile by trailing 12-month return; Jegadeesh and Titman 1993 extended to multi-asset); trend-following via EWMA crossover (long when 63-day EWMA exceeds 252-day EWMA, equal-weight among trending positions; standard managed-futures implementation); and risk parity (weights inversely proportional to trailing 63-day realised volatility, rebalanced daily, no leverage).

18.1 Results

Table 13 reports the full comparison. All four standard strategies underperformed the episode-validated equal-weight baseline (Sharpe 1.317) in 2025. TSMOM produced 8.45% / Sharpe

0.822; XSMOM 11.24% / Sharpe 0.757; trend-following 10.10% / Sharpe 0.956; risk parity 10.06% / Sharpe 1.555. All three signal-based strategies (TSMOM, XSMOM, trend-following) were active on every one of the 250 evaluation days, since their signals are continuous rather than episodic. Risk parity achieved the highest Sharpe among the standard strategies at 1.555, close to but below the CPE cross-sectional result (1.613). The CPE cross-sectional top-quartile strategy (24.58% / Sharpe 1.613, pct_exceeding 2.2%) produced the highest Sharpe in the table. The CPE five-sleeve hold-to-horizon strategy (18.77% / Sharpe 1.224, pct_exceeding 1.8%) exceeded SPY buy-and-hold (18.01% / 0.955) on both return and risk-adjusted basis while operating on only 15 of 250 days.

Table 13. CPE strategies vs standard quantitative benchmarks, 2025 (250 trading days, 61-target episode-validated universe, long-only, zero transaction costs, zero risk-free rate). Active days = days with non-uniform weights. pct_exc = randomisation test pct_exceeding; -- indicates test not run.

SPY buy-and-hold: return 18.01%, vol 19.52%, Sharpe 0.955, active days 250/250, pct_exc --.

Equal-weight (61-target episode-validated universe): return 14.45%, vol 10.81%, Sharpe 1.317, active days 0/250 (no signal), pct_exc --.

TSMOM (12-month lookahead): return 8.45%, vol 10.69%, Sharpe 0.822, active days 250/250, pct_exc --.

XSMOM top-quartile (12-month lookahead): return 11.24%, vol 15.89%, Sharpe 0.757, active days 250/250, pct_exc --.

Trend-following EWMA (63d/252d): return 10.10%, vol 10.80%, Sharpe 0.956, active days 250/250, pct_exc --.

Risk parity (equal-vol, 63d): return 10.06%, vol 6.37%, Sharpe 1.555, active days 250/250, pct_exc --.

CPE cross-sectional top-quartile (Structure B): return 24.58%, vol 14.42%, Sharpe 1.613, active days 30/250, pct_exc 2.2%.

CPE five-sleeve hold-to-horizon (corrected weights): return 18.77%, vol 15.15%, Sharpe 1.224, active days 15/250, pct_exc 1.8%.

18.2 Assessment

The 2025 environment penalised standard momentum and trend strategies. TSMOM, XSMOM, and trend-following all underperformed the equal-weight baseline and all underperformed SPY on Sharpe. This is consistent with 2025 being a reversal-dominated year (sharp drawdown in April followed by an equally sharp recovery), which is structurally adverse for 12-month momentum and EWMA crossover strategies. Their underperformance in 2025 reflects the market environment as much as the strategies' own properties; a multi-year comparison would be required to draw conclusions about their relative quality.

Risk parity is the most relevant standard benchmark for CPE. Risk parity achieved Sharpe 1.555 with no signal and no randomisation test required — purely by vol-weighting the same 61-target universe. The CPE cross-sectional strategy beat it by 0.058 Sharpe units (1.613 vs 1.555) and by 14.52 percentage points in return (24.58% vs 10.06%). Whether this margin is real or a 2025-specific result is the open question. The CPE strategy operated on 30 of 250 days; risk parity operated on all 250. The CPE signal contributed over a narrow set of days and was inactive for 88% of the year. Risk parity's lower return in 2025 reflects its lower allocation to equities (the dominant return driver in 2025) in favour of lower-volatility assets; it is designed to be a lower-return, lower-vol strategy rather than a return-maximiser.

Universe selection explains more of the CPE advantage than signal construction. The episode-validated equal-weight baseline (Sharpe 1.317) already beats TSMOM (0.822), XSMOM

(0.757), and trend-following (0.956) without any signal. The signal's contribution in Structure B is the additional 0.296 Sharpe units above the equal-weight baseline (1.613 vs 1.317). Filtering the 161-instrument universe down to 61 targets with at least 3 independent historical episodes produces a better-diversified portfolio than these standard strategies achieved on the same universe, independent of any signal. This has a practical implication: the episode-validation filter may be valuable as a portfolio construction tool (universe selection) even in contexts where the CPE signal itself is not used for active tilting.

The comparison is one year and should not be overstated. All results above are from a single evaluation year in which the specific market structure (April reversal, equity-led recovery) happened to favour the CPE strategy's regime-specific signal and penalise the continuous-signal benchmark strategies. The CPE cross-sectional result (Sharpe 1.613, pct_exceeding 2.2%) is the strongest single result in this paper, but it rests on 30 firing days in one calendar year. A meaningful comparison with standard quant strategies requires at minimum three to five years across different market regimes, including at least one equity bear market, one low-volatility bull market, and one rate-driven drawdown environment — none of which are represented in the 2025 evaluation window alone.

19. Next Steps and Open Questions

Sections 14–18 completed the three open scopes identified at the end of the original investigation (Section 14.6) and produced the paper's first statistically significant results. Scopes 19.2–19.5 have since been completed and are reported in Section 20; this section preserves the original pre-specification of each scope for the record, updated to note where results are now available.

19.1 Immediate Priority: Full-Year 2026 Evaluation

The single most informative next test is a full calendar-year 2026 out-of-sample run using the corrected pipeline (economic prior, episode-conviction sizing, corrected neutral weights) with no parameter changes of any kind. The partial 2026 window reported in Sections 15.4, 16.3, and 17.3 covers approximately five months and is uninformative by construction — the strategy fires only on rare, multi-decade-validated conditioning events, none of which occurred in the first five months of 2026. As of the time of writing (June 2026), approximately six months of 2026 data remain before a full calendar year is available. The partial 2026 window already reported (Sections 15.4, 16.3, 17.3) shows zero signal fires, which is consistent with the framework's design — no genuine volatility-spike-then-recovery episode occurred in the first five months of 2026 — but is by construction uninformative about signal quality. A full calendar year provides a meaningful second data point on whether the 2025 significance (pct_exceeding 1.8–2.2%) replicates across a distinct market regime. Critically, Section 20's data-sufficiency finding means this test is interpreted differently depending on whether the VIXM+VIXY conditioning event occurs in 2026: if it fires and resolves correctly, the stored episode count rises to 5 and the framework's track record becomes somewhat more distinguishable from chance; if 2026 passes without a fire, it provides no additional information either way. This test requires no new code; it should be the first thing run when 2026 year-end data is available, expected around January 2027.

19.2 Disentangle Base Allocation from Signal Quality

Section 16.4 identified a tension that remains unresolved: the 5-sleeve book has a validated signal but a structurally poor base allocation (30% BTC-USD and 24% Gold drag performance

in equity-led markets), while the 9-sleeve book has a better base allocation but sector signals that actively hurt performance in both evaluation years. No tested configuration simultaneously achieves a sound base allocation and a validated tilt signal. The specific test needed is: run the corrected-weight 5-sleeve specification (Sharpe 1.224, pct_exceeding 1.8%) with the Crypto sleeve weight capped at 15% and the Gold sleeve weight capped at 20%, reallocating the surplus to Equities and Bonds proportionally. This is a single structural change motivated by a pre-specified concern — not an optimisation. This test has since been completed (Section 20.1): the capped weights produced a hold-to-horizon Sharpe of 1.464 (pct_exceeding 4.6%, 1,000 reps, 22 holds shuffled), remaining statistically significant. The randomisation test tightened slightly from 1.8% to 4.6% under the cap, confirming the signal's significance is not driven by elevated Crypto or Gold beta exposure.

19.3 Transaction Cost Break-Even Analysis

All results in this paper assume zero transaction costs. Structure B (cross-sectional top-quartile, Sharpe 1.613) trades on 30 of 250 days; the 5-sleeve hold-to-horizon strategy trades on 15 of 250 days. The break-even one-way transaction cost at which each strategy's edge disappears is a straightforward calculation that has not been reported anywhere in this series. For a quant allocator evaluating the practical value of the framework, this is the first number they will ask for. This analysis has since been completed (Section 20.1): the 5-sleeve hold-to-horizon strategy has a break-even of 10.1 bps per one-way leg (22 legs), comfortably above realistic ETF transaction costs of 1–5 bps. The cross-sectional Structure B break-even ranges from 18.8 to 56.3 bps per leg depending on turnover assumptions. Both strategies survive realistic cost assumptions; the 2022 OOS result (Section 20.3) does not (break-even 0.6 bps, erased at any realistic cost).

19.4 Validate Short-Conditioning-Window Sector Signals Independently

The JNK/HYG → XLY/XLI/XLP signals ($\tau_{\text{past}} = 5$, 45–81 independent training episodes, 80–87% hit rates) are the episode-conviction screen's strongest signals by episode count — dramatically stronger than the VIXM+VIXY → SPY signal (6 episodes) that produced the only validated result to date. Yet when added to the tradeable universe in Section 16.2 they reversed the HTH randomisation result from 1.8% exceeding to 80.5% exceeding. The failure mode is likely specific to the 2025 market environment rather than a permanent property of these signals, but that cannot be distinguished from a single year's data. This test has since been completed (Section 20.1, Scope 19.4): a multi-year walk-forward (2020–2024) on these signals alone in an isolated sector-only sleeve book found mixed results — 2 of 5 years positive, 1 of 5 statistically significant (2022, 0.7%, in the downside-protection direction). The verdict is that the sector signals are not ready for re-inclusion; the one significant year reflects bear-market loss mitigation rather than directional return enhancement, and the signals failed in both 2020 (99.2%) and 2023 (95.6%). A $\tau_{\text{past}} \geq 63$ variant is recommended before further consideration.

19.5 Multi-Year Walk-Forward on the Validated VIXM+VIXY → SPY Signal

The VIXM+VIXY → SPY joint configuration is the only signal validated to date. It has 6 independent training-period episodes (2011, 2018–2019, 2020, 2022, 2024) and fired correctly in April 2025. The single-asset paper's Section 8.12 demonstrated that a multi-decade walk-forward can reverse or confirm conclusions reached on one year alone. A walk-forward test of this specific configuration across 2015–2024 (using expanding training windows, never touching 2025 data) would establish whether the 2025 significance is specific to that year's market structure or reflects a genuine multi-regime pattern. This test has since been completed as part

of Section 20's proper OOS walk-forward, which re-derives the full pipeline for each evaluation year from 2010 through 2024. The results are more nuanced than either a simple confirmation or refutation: the three statistically significant OOS years (2011, 2020, 2022) were driven by rate-index→bond and commodity→gold channels rather than the validated VIXM+VIXY→SPY mechanism. The episode floor sensitivity test (Section 20.6) further established that the stored n_episodes for VIXM+VIXY→SPY configurations is 4 (not 6 as stated in earlier runtime diagnostics), reflecting an inconsistency between the joint screen's stored episode count and the backtest engine's runtime recomputation. The vol→equity channel fired correctly on the August 2024 carry-trade unwind but did not produce a significant result given the randomisation test's limited power at 4 hold events. The full interpretation is given in Section 20.5.

19.6 Longer-Term Research Directions

Three further directions are identified as valuable but lower priority than the five tests above, since each requires either new methodology or a substantially longer data window. All three are also downstream of the overarching data-sufficiency question established in Section 20: the framework's validated channel currently rests on four independent training-period episodes, and no methodological refinement can substitute for the additional episodes that only time and future market stress events can provide. The directions below are therefore presented as the correct next steps given the current evidence, not as paths that would independently resolve the question of whether the CPE framework has genuine predictive content:

- Conditioning threshold non-stationarity: Section 5 diagnosed that structurally decaying instruments produce frozen thresholds that can never be re-cleared. A principled fix — computing quantiles on detrended increments rather than raw trailing returns for instruments with documented roll decay — would allow re-inclusion of the VIX-complex ETPs as predictors without the decay confound. This is a methodological change requiring careful validation before being applied.
- Live trading with timestamps: the strongest possible evidence for the framework's signal quality is a live, time-stamped trading record in which positions are opened before the forward return is known. The partial 2026 window provides the beginning of such a record (zero fires, consistent with the strategy's design). A meaningful accumulation requires at minimum three to five additional conditioning events firing and resolving in real time — not one or two, because the data-sufficiency analysis in Section 20 establishes that four stored episodes is the minimum that produces a selective screen, and a live record needs to add meaningfully above that baseline to begin distinguishing genuine skill from a well-supported coincidence. At the historical frequency of qualifying volatility episodes (roughly one per two to three years), this implies a live trading record of at least a decade before the data-sufficiency question can be addressed through live performance.
- Generalisation beyond the CPE screen's current 161-instrument universe: the Atlas and this paper both restrict to instruments available in the author's live dashboard. Extending the predictor and target universes — particularly to include macro factors (yield curve shape, credit spreads, PMI diffusion indices) as predictors — would test whether the validated VIXM+VIXY → SPY pattern is a specific instance of a broader volatility-regime-to-equity-recovery relationship that macro predictors might identify earlier or more reliably.

20. Proper Out-of-Sample Walk-Forward Test

Sections 19.2–19.5 completed the scopes identified for the validated VIXM+VIXY→SPY signal. This section reports a more fundamental pre-publication verification: a proper out-of-sample walk-forward test in which the full CPE pipeline is re-derived from pre-evaluation data each year, the framework’s signal quality is assessed across multiple channels through a diagnostic tracing of which configurations actually fired, and a single-screen continuous evaluation tests whether the framework generalises across an eleven-year window. Together these three tests resolve the paper’s central open question: does the 2025 significance hold up when the configuration set itself is derived from data the evaluation year never touched?

20.1 Design: Annual Expanding-Window Pipeline Re-Derivation

For each evaluation year Y in $\{2010, 2011, \dots, 2024\}$, the full pairwise CPE screen, greedy joint screen, and hold-to-horizon backtest were re-derived using only data through $(Y-1)$ -12-31. No configuration, threshold, or conviction weight from any prior year’s screen was carried forward. The five-sleeve book with capped neutral weights (Crypto $\leq 15\%$, Gold $\leq 20\%$, Section 19.2) was used throughout. The same coarser parameter grid as the paper’s main evaluation ($\tau_p \in \{5, 21, 63, 126, 252\}$, $\tau_e \in \{21, 63, 126, 252\}$) was applied at each re-derivation step, and the same economic prior, episode-conviction sizing, and MIN_TRAIN_OBS = 500 floor were enforced. Results were evaluated year by year; the randomisation test was applied to each year with ≥ 3 hold events independently.

20.2 Results: Annual Walk-Forward (2010–2024)

Table 14 reports the full annual walk-forward. Three of eleven testable years produced statistically significant results: 2011 (pct_exceeding 5.4%, 3 holds), 2020 (6.6%, 4 holds), and 2022 (2.2%, 4 holds). The data availability check confirmed VIXM and VIXY had zero training observations through 2010-12-31 and only 251 observations through 2011-12-31, meaning the validated vol→equity channel was structurally unavailable for evaluation years 2010–2012.

Year	HTH ret%	HTH Sharpe	Bench Sharpe	SPY ret%	Holds	Pct exc	VV active	Sig?
2010	13.71%	1.826	1.921	13.14%	4	35.3%	0	no
2011	7.67%	0.640	2.149	0.85%	3	5.4%	0	YES
2012	6.82%	0.843	1.169	14.17%	7	23.2%	0	no
2013	−5.71%	−0.877	−0.485	29.00%	16	97.5%	0	no
2014	17.88%	3.184	2.634	14.56%	10	95.3%	0	no
2015	2.49%	0.327	0.039	1.29%	8	98.9%	1	no
2016	11.72%	1.625	1.555	13.59%	7	24.2%	0	no
2017	60.85%	3.623	3.934	20.78%	5	83.8%	0	no
2018	−19.38%	−1.277	−1.380	−5.25%	2	n/a	0	—
2019	32.70%	2.495	2.495	31.09%	1	n/a	0	—

2020	60.01%	2.455	2.244	17.24%	4	6.6%	0	YES
2021	16.88%	1.184	1.184	30.51%	0	n/a	0	—
2022	-24.19%	-1.604	-1.529	-18.65%	4	2.2%	3	YES
2023	29.33%	2.366	2.534	26.71%	1	n/a	4	—
2024	29.20%	2.254	2.246	25.59%	4	51.5%	4	no
2025*	18.77%	1.224	1.089	18.01%	11	1.8%	—	YES

*Table 14. Annual walk-forward results, 2010–2024 (full pipeline re-derived each year). ‘VV active’ = number of VIXM+VIXY→SPY configs with episode_conviction > 0 at that year’s training cutoff. Dashes (—) in Sig? column indicate fewer than 3 hold events (randomisation test not applicable). * 2025 result from paper Section 16.1, trained on ≤2024-12-31 frozen screen, reported for reference.*

A strong inverse relationship between hold count and outcome quality is observed across the full table. Years generating 3–4 hold events produced 3 of 4 significant results (2011, 2020, 2022, and a borderline 2010 at 35.3%), while years generating 5–10 holds produced no significant results, and years generating 11 or more holds (2013: 97.5%, 2014: 95.3%, 2015: 98.9%) were actively worse than random timing of the same exposures. This pattern is consistent with the saturation mechanism documented throughout this paper: when the freshly-derived screen fires frequently, it has selected spurious-but-recurring relationships that the episode-conviction floor of 3 does not fully exclude at short training windows.

20.3 Diagnostic: Which Configurations Actually Fired?

The three significant years were subjected to full hold-event tracing to determine whether they share the validated vol→equity channel or represent independent economic mechanisms. Table 15 summarises the findings.

Year	Sig?	Holds	Primary channel	Vol-driven	VV-driven
2011	YES	3	rate_index → bond_credit	0%	0%
2020	YES	4	commodity/gold → gold	0%	0%
2022	YES	4	rate_index → bond_credit	0%	0%
2024	no	4	vol_etp → equity	100%	100%
2025*	YES	11	vol_etp → equity	100%	27%

*Table 15. Economic channel classification of hold events in significant and comparison years. ‘VV-driven’ = fraction of holds where both VIXM and VIXY appear in the predictor set. * 2025 uses the paper’s own frozen screen; channel classification derived from diagnostic tracing.*

The three OOS-significant years operated on entirely different channels from the paper’s validated VIXM+VIXY→SPY mechanism. In 2011, rate indices (^TNX, ^TYX, ^FVX) predicting TLT bearish drove all three holds, coinciding with the European sovereign debt crisis and US

debt ceiling shock that produced sharp long-rate movements. In 2020, palladium and platinum futures (PPLT, PALL) predicting gold bearish drove the majority of holds, alongside rate indices predicting TLT, consistent with the COVID-driven collapse in industrial metals preceding a gold dislocation. In 2022, rate indices and HYG/JNK jointly predicting TLT across 35 hold events produced a statistically significant result (pct_exceeding 2.2%) but a gross return advantage of only 0.39 percentage points over the benchmark — erased entirely at any realistic transaction cost (break-even: 0.6 bps per one-way leg; TLT’s realistic spread of ~0.8 bps produces a net disadvantage).

The vol→equity channel appears in the OOS screen only from 2022 onwards, when VIXM and VIXY have accumulated sufficient training history for the episode filter to pass. The configuration fired correctly on the August 2024 carry-trade unwind (4 holds, all VIXM+VIXY→SPY, pct_exceeding 51.5% — not significant given the tight null distribution at 4 holds) and produced the paper’s primary 2025 result. The three OOS-significant years therefore do not constitute independent replications of the specific mechanism validated in 2025; they demonstrate that the CPE framework identifies genuine tail co-movements across multiple economic channels, but that different channels produce significant results in different regimes.

20.4 Single-Screen Continuous Evaluation (2015–2025)

To address the low-power problem of year-by-year testing and the confound of annual screen re-derivation, a single frozen screen was trained on all data through 2014-12-31 and evaluated continuously across 2015-01-01 to 2025-12-31 (2,766 trading days). At the 2014-12-31 training cutoff, VIXM and VIXY each had 1,005 training observations — sufficient for the episode-conviction filter to pass at exactly 3 episodes per the stored n_episodes count (conv = 1.099, CPE = 0.846, tau_f = 63), the minimum floor. Note that the 2024-trained screen stores 4 episodes for the same configuration; the difference reflects additional episode history accumulated between 2014 and 2024. The capped neutral weights (Equities 33.5%, Gold 20.0%, Bonds 39.3%, Crypto 3.6%, FX 3.6%) reflected the relatively short Crypto history available by 2014.

The strategy generated 785 hold events across the eleven-year evaluation period (421 Gold, 305 Equities, 59 Bonds), averaging 71 holds per year and peaking at 125 in 2015. The hold-to-horizon backtest produced a cumulative return of 171.57% (Sharpe 0.904) versus the no-tilt benchmark of 193.29% (Sharpe 1.095). A single pooled randomisation test across all 785 hold events found pct_exceeding = 72.7% (null mean 0.908, null std 0.007, 1,000 repetitions) — the strategy performed worse than a typical random permutation of the same exposures.

The 785 hold events are themselves the diagnostic result. With 251 joint configs and 16 targeting the five tradeable sleeves, the 2014-trained screen generates configurations that fire nearly continuously rather than selectively. Despite the VIXM+VIXY→SPY configuration being present and active (episode_conviction > 0), it contributes only a small fraction of total hold events, swamped by Gold and Equities configs that fire spuriously across the full eleven-year window. This is the same saturation mechanism documented in Section 8 and consistently observed across all OOS tests: the episode-conviction floor of 3 independent episodes at a 2014 training cutoff is insufficient to prevent non-selective firing when many configurations are present with thin but non-zero historical support.

Specification	Train cutoff	Eval period	Total holds	HTH Sharpe	Bench Sharpe	Pct exceeding
Annual re-derivation (Table 14)	Expanding	2010–2024	3–16/year	Varies	Varies	3/11 sig years

Single frozen screen	2014-12-31	2015–2025	785	0.904	1.095	72.7% (not sig)
Paper primary result (Sec 16.1)	2024-12-31	2025 only	11	1.224	1.089	1.8% (sig)

Table 16. Comparison of OOS test designs. All use the five-sleeve hold-to-horizon mechanism with capped neutral weights.

20.5 Interpretation: What the OOS Tests Collectively Establish

Across all three OOS designs, the evidence converges on four findings that are now firmly established.

The 2025 result is genuine but training-window dependent. The paper's primary finding (pct_exceeding 1.8%, Section 16.1) requires a screen trained through 2024-12-31, which gives VIXM+VIXY six confirmed independent episodes and produces eleven selective hold events. Trained through 2014-12-31, the same framework fires 785 holds and underperforms its benchmark by 0.191 Sharpe units. The framework's signal quality is directly proportional to the episode history available at training time for the specific channel being tested.

The validated channel requires a minimum of four stored independent episodes to dominate a selective screen. With three episodes (2014 cutoff), VIXM+VIXY→SPY survives the episode-conviction filter but is overwhelmed by other configurations. With four episodes (2024 cutoff, tau_f = 63 configurations), it produces the selective 2025 screen. Section 20.6 establishes that the stored n_episodes for these configurations is 4, correcting an earlier runtime diagnostic that reported 6; the four genuine independent episodes are the 2011 European debt crisis, 2018–2019 growth scare, 2020 COVID crash, and 2024 carry-trade unwind. This is a concrete, quantified finding about the framework's data requirements: the vol→equity channel became reliably detectable only after the 2022 and 2024 volatility episodes extended the training history beyond five independent occurrences.

The episode-conviction floor of three is necessary but not sufficient. The minimum floor prevents the worst artifacts — at a 2024 training cutoff, it zeroed out the configurations responsible for the 785-hold saturation — but at a 2014 training cutoff the same floor permits configurations that fire continuously despite having genuine but shallow historical support. Raising the minimum episode floor to five or six, applied as a hard screen filter at joint-screen construction time, appears to be the natural next correction. Section 20.6 tests this directly and finds a prior step is required: the stored n_episodes values in the joint screen and the backtest engine's runtime episode recomputation are currently inconsistent, meaning the floor filter must first be applied to a rebuilt screen with consistent episode counting before the recommendation can be properly evaluated.

The framework functions as a rare-event detector, not a general timing system. All positive OOS results — 2011, 2020, 2022 from the annual walk-forward, and 2025 from the paper's primary result — occurred in years corresponding to identifiable market stress or regime-transition events (European debt and US debt ceiling crisis, COVID crash, rate-shock bear market, tariff shock). The framework is silent in ordinary market environments and fires only when a conditioning event with genuine multi-regime historical support occurs. This episodic character, while limiting the strategy's activity level, is consistent with the CPE framework's original design as a detector of tail co-movement in complex systems — events that are precisely defined by their rarity, not their frequency.

These four findings do not overturn the paper’s primary result. They contextualise it: the 2025 signal (pct_exceeding 1.8%) is the first year in which the validated channel had accumulated sufficient independent episode support to produce a genuinely selective screen, and the OOS tests confirm that the framework correctly abstains in the years when that support is absent. The appropriate summary is that the CPE framework has produced one robustly validated signal in one specific market channel, contingent on that channel’s predictors having cleared a minimum of approximately six independent historical episodes in training data — a condition first met by the end of 2024.

20.6 Episode Floor Sensitivity Test and a Methodological Finding

Section 20.5 identified a concrete actionable recommendation: raising the minimum episode floor from 3 to 5–6, applied as a hard screen filter at joint-screen construction time. This section tests that recommendation across three combinations and reports an unexpected but important methodological finding that changes the interpretation.

The three combinations tested were: (A) the 2024-trained screen with episode floor = 5, evaluated over 2025; (B) the 2024-trained screen with episode floor = 6, evaluated over 2025; and (C) the 2014-trained screen with episode floor = 5, evaluated over 2015–2025 (the single-screen continuous evaluation of Section 20.4). Table 17 summarises the results alongside the reference results from Sections 16.1 and 20.4.

Specification	Floor	Configs	Holds	HTH Sharpe	Bench Sharpe	Pct exceeding
2024-trained, floor=3 (paper Sec 16.1, 2025)	3	655	11	1.224	1.089	1.8% (sig)
2024-trained, floor=5 (Test A, 2025)	5	146	1	1.468	1.369	n/a (<3 holds)
2024-trained, floor=6 (Test B, 2025)	6	128	1	1.468	1.369	n/a (<3 holds)
2014-trained, floor=3 (Sec 20.4, 2015–2025)	3	251	785	0.904	1.095	72.7% (not sig)
2014-trained, floor=5 (Test C, 2015–2025)	5	44	277	1.050	1.095	21.6% (not sig)

Table 17. Episode floor sensitivity results. ‘Configs’ = joint configs surviving the floor filter targeting the five tradeable sleeves. All runs use the hold-to-horizon mechanism with capped neutral weights.

The primary finding from Tests A and B is not what was anticipated. Inspecting the stored `n_episodes` values in `joint_cpe_results.parquet` directly reveals that the `VIXM+VIXY→SPY` configurations at `tau_f = 63` — the ones that drove the paper’s 2025 result — have `n_episodes = 4` in the stored joint screen, not 6 as reported in the paper’s runtime diagnostics. At floor = 5, these configurations are therefore removed, leaving only 1 hold event in 2025 (from a different Equities configuration with `n_episodes = 6–10`). This 1 hold is untestable by randomisation. The

VIXM+VIXY configurations at $\tau_f = 252$ and $\tau_f = 300$ have $n_{\text{episodes}} = 1$ and $\text{episode_conviction} = 0$ throughout, meaning they were already inactive.

This discrepancy between stored and runtime episode counts is itself the methodological finding. The joint screen's n_{episodes} column was computed during the greedy search using the episode-clustering implementation current at screen-build time. The backtest engine's runtime episode recomputation — which the paper's Section 14.3 documents was corrected for a forward/trailing return direction bug — produces different counts. After the correction, the runtime recomputation finds 4 genuinely distinct temporal episodes for the $\tau_f = 63$ VIXM+VIXY→SPY configurations (2011, 2018–2019, 2020, 2024), not 6. The paper's earlier statement that this channel has “6 independent training-period episodes” reflected a diagnostic that counted episode groups differently from both the stored screen and the corrected runtime engine. The correct count, using the episode-clustering algorithm consistently applied, is 4.

This has two consequences. First, the recommendation to raise the floor to 5–6 would, if applied to the existing screen, eliminate the validated VIXM+VIXY→SPY channel entirely. The recommendation therefore requires a preceding step: rebuilding the joint screen with the corrected episode-counting algorithm applied consistently, so that the stored n_{episodes} column reflects the same computation as the runtime engine. Only then can a floor filter be applied without inadvertently removing channels whose true episode count clears the floor. Second, the paper's claim about the number of independent episodes should be corrected: the VIXM+VIXY→SPY channel at $\tau_f = 63$ has 4 stored episodes ($n_{\text{episodes}} = 4$, $\text{episode_conviction} = 1.386$) under the current screen, not 6.

Test C (2014-trained screen, floor = 5) reduced the hold count from 785 to 277 — a 65% reduction — but saturation persisted. The surviving Equities configuration has $n_{\text{episodes}} = 30$, firing 227 times across the 11-year window. This reveals a second limitation of the episode floor approach as currently implemented: a configuration with many episodes can still fire at very high frequency if its conditioning event is common, because the episode count measures temporal separation of firing clusters but does not constrain within-cluster firing density. At 277 pooled holds across 11 years, the randomisation test (pct_exceeding 21.6%) confirms the strategy is not statistically distinguishable from random timing of the same exposures.

Three specific methodological corrections follow from this section's findings, each well-defined and testable in a future iteration: (1) rebuild the joint screen with the episode-counting algorithm applied identically in the greedy search and the backtest engine, so that stored n_{episodes} and runtime recomputed values agree; (2) apply the episode floor filter to the rebuilt screen, where the VIXM+VIXY→SPY channel at $\tau_f = 63$ should correctly show $n_{\text{episodes}} = 4$ and therefore survive floor = 3 but not floor = 5; and (3) add a maximum firing frequency constraint alongside the episode floor — for instance, requiring that a configuration fires on fewer than a specified number of days per evaluation year on average across its training history — to prevent high-episode configurations from contributing to saturation through continuous within-cluster firing. None of these corrections is attempted here; reporting the diagnosis precisely and leaving the fix to a future study with its own evaluation window is consistent with this paper's discipline throughout.

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A note on scope

This manuscript characterises a research prototype's historical out-of-sample performance over a single evaluation year and is not investment advice. Nothing in it constitutes a recommendation to transact in any instrument. Past performance carries no guarantee of future results, and the omission of transaction costs and capital constraints beyond a single notional book means the reported figures should not be read as deployable net returns.